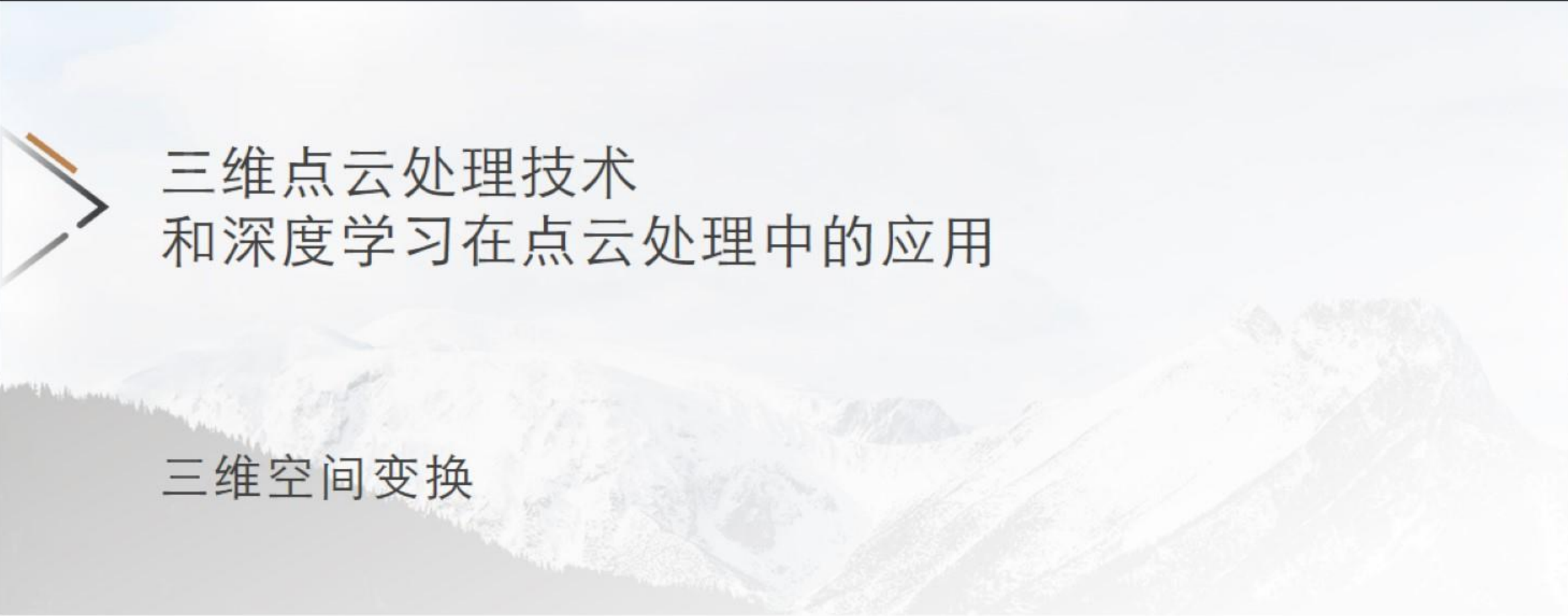




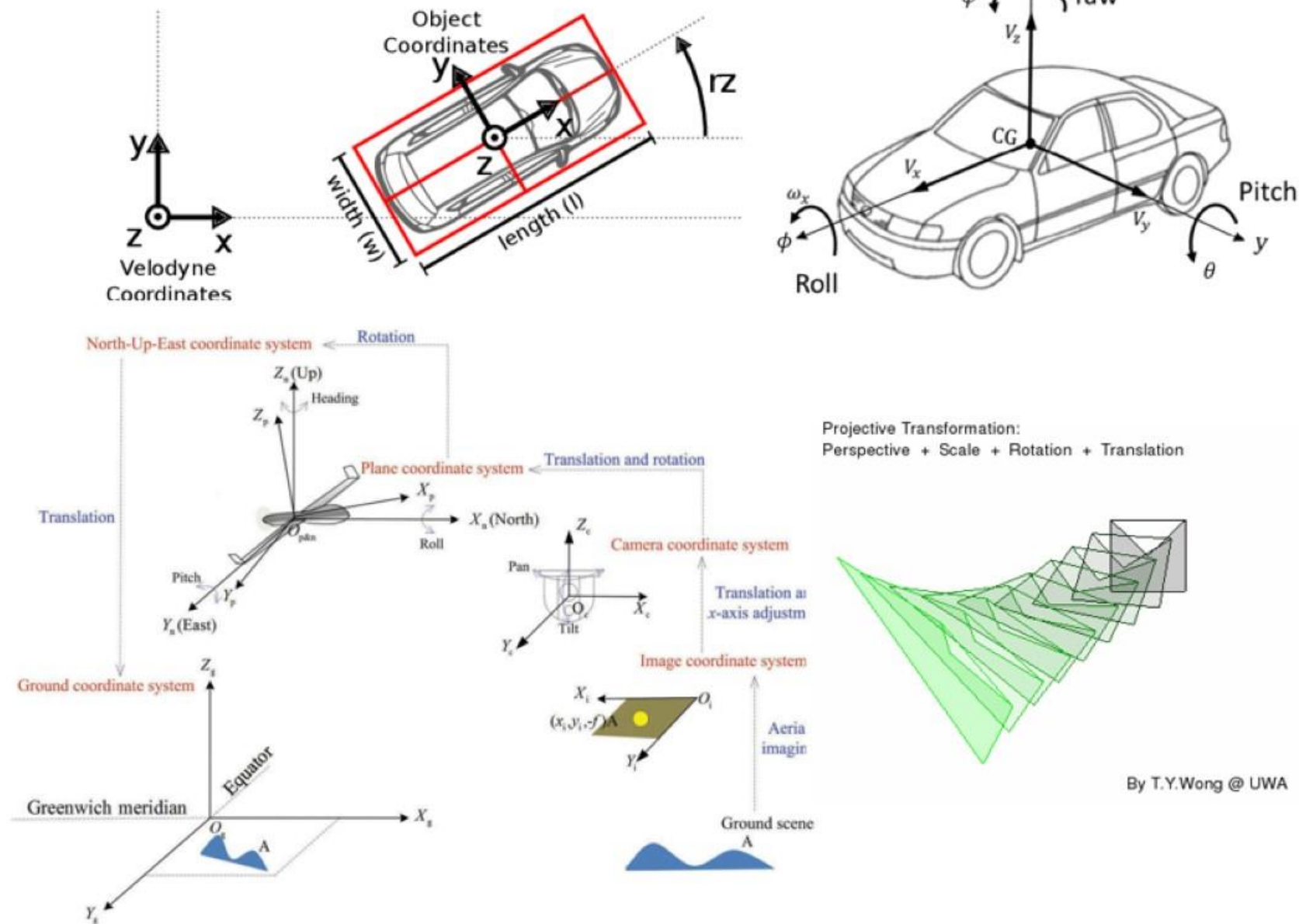
三维点云处理技术 和深度学习在点云处理中的应用

三维空间变换

索传哲

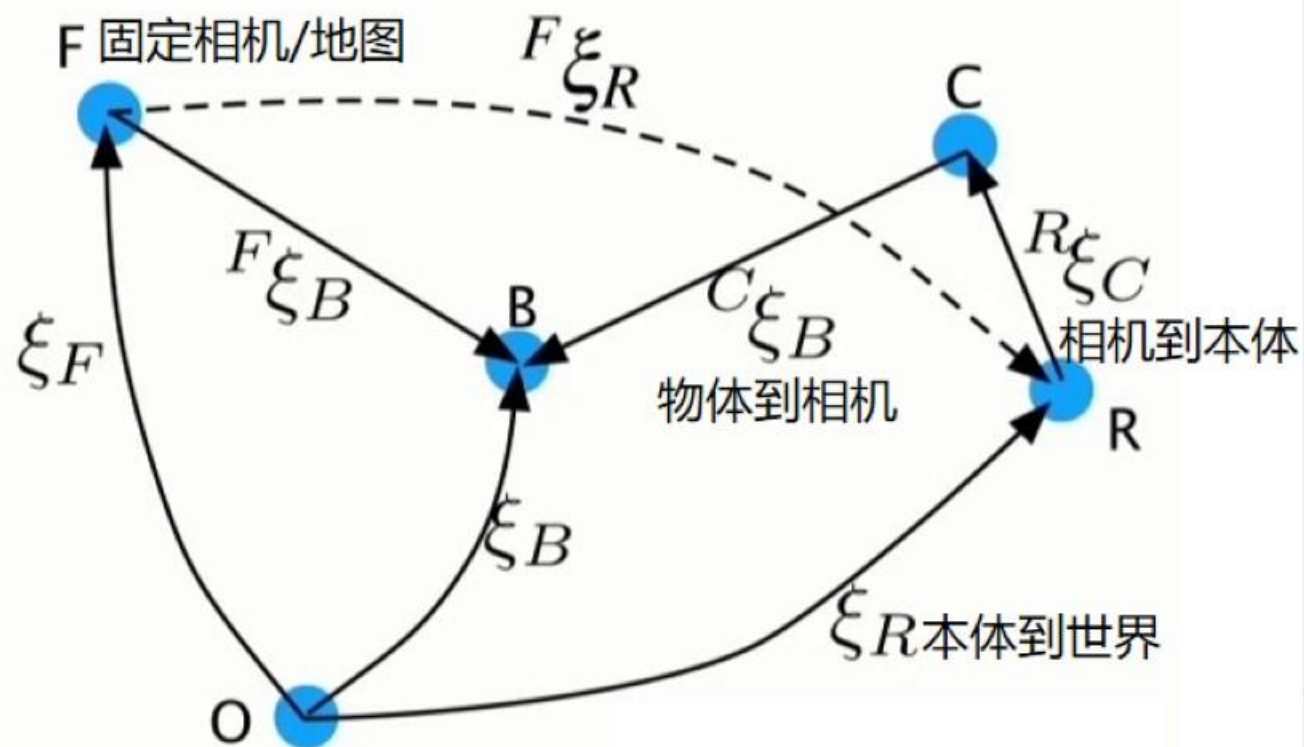
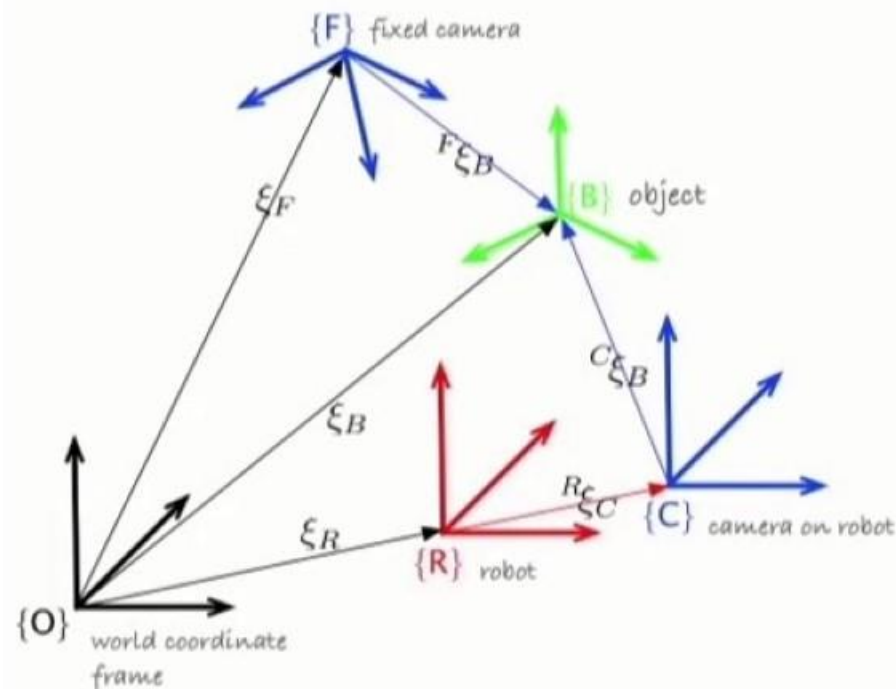
三维空间变换

- 欧式坐标系
- 刚体运动变换
- 三维空间变换
(射影、仿射、比例、欧式)



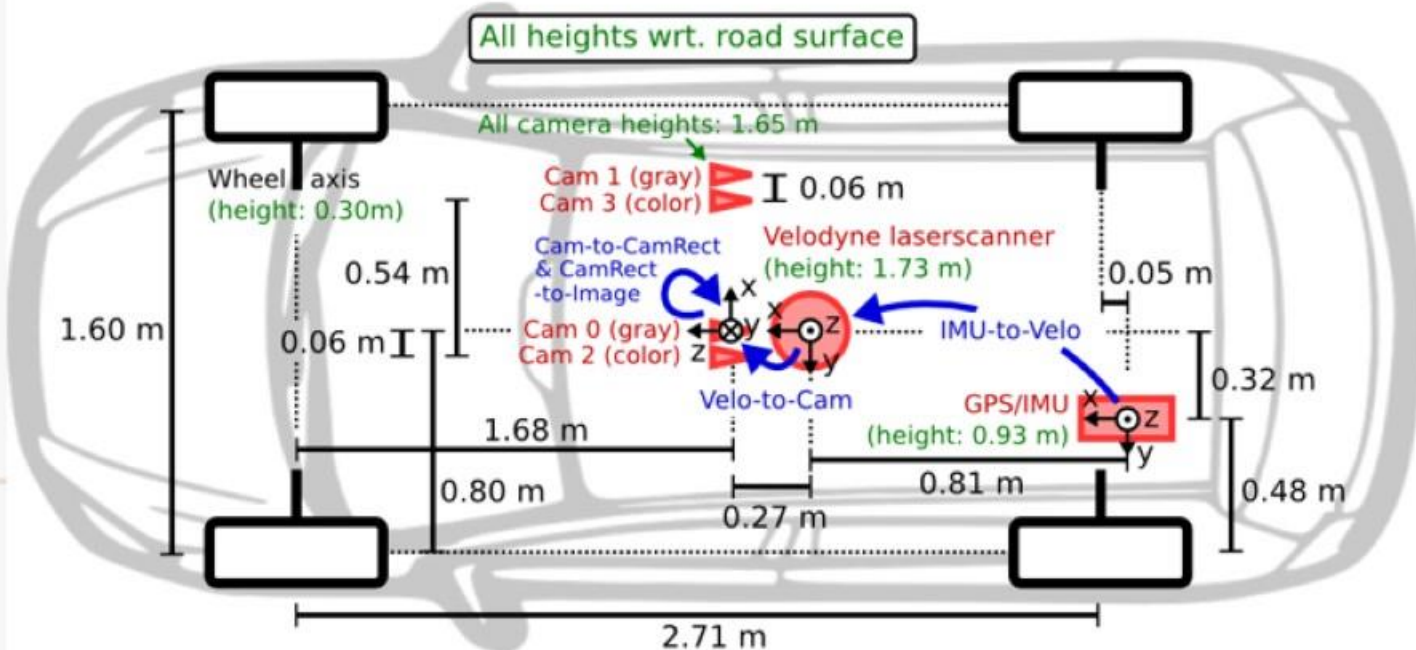
三维空间变换

➤ 欧式坐标系:



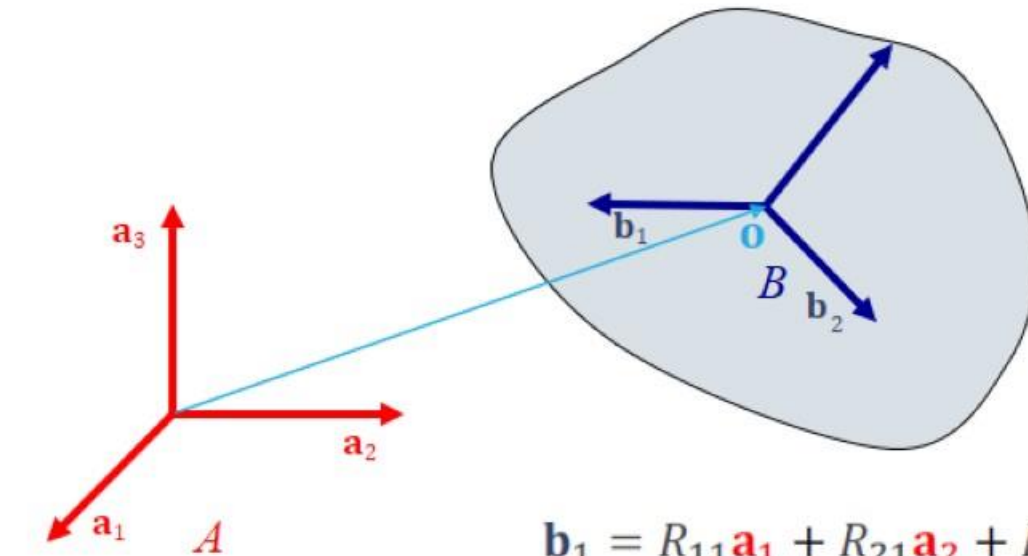
三维空间变换

➤ 欧式坐标系:



三维空间变换

- 刚体运动变换:
 - 刚体位姿表示



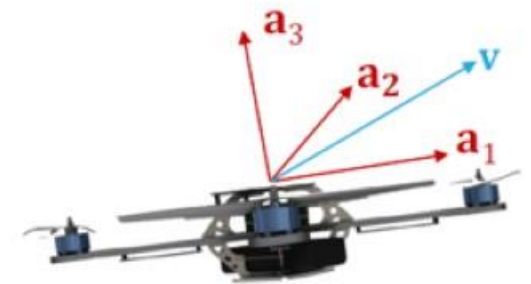
$$\mathbf{o} = [o_1 \ o_2 \ o_3]$$

$$\mathbf{b}_1 = R_{11}\mathbf{a}_1 + R_{21}\mathbf{a}_2 + R_{31}\mathbf{a}_3$$

$$\mathbf{b}_2 = R_{12}\mathbf{a}_1 + R_{22}\mathbf{a}_2 + R_{32}\mathbf{a}_3$$

$$\mathbf{b}_3 = R_{13}\mathbf{a}_1 + R_{23}\mathbf{a}_2 + R_{33}\mathbf{a}_3$$

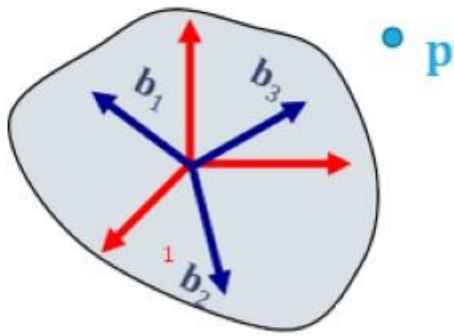
$$\mathbf{v} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + v_3\mathbf{a}_3$$



三维空间变换

➤ 刚体运动变换:

- 旋转矩阵 (Rotation Matrix)



$$\mathbf{b}_1 = R_{11}\mathbf{a}_1 + R_{21}\mathbf{a}_2 + R_{31}\mathbf{a}_3$$

$$\mathbf{b}_2 = R_{12}\mathbf{a}_1 + R_{22}\mathbf{a}_2 + R_{32}\mathbf{a}_3$$

$$\mathbf{b}_3 = R_{13}\mathbf{a}_1 + R_{23}\mathbf{a}_2 + R_{33}\mathbf{a}_3$$

$$\boxed{\mathbf{R}_B^A} = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1^T \mathbf{a}_1 & \mathbf{b}_2^T \mathbf{a}_1 & \mathbf{b}_3^T \mathbf{a}_1 \\ \mathbf{b}_1^T \mathbf{a}_2 & \mathbf{b}_2^T \mathbf{a}_2 & \mathbf{b}_3^T \mathbf{a}_2 \\ \mathbf{b}_1^T \mathbf{a}_3 & \mathbf{b}_2^T \mathbf{a}_3 & \mathbf{b}_3^T \mathbf{a}_3 \end{bmatrix}$$

$$\mathbf{p}_A = p_{1b}\mathbf{b}_1 + p_{2b}\mathbf{b}_2 + p_{3b}\mathbf{b}_3 = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} \mathbf{p}_B \quad \mathbf{p}_A = \mathbf{R}_B^A \mathbf{p}_B$$

B的基向量在A中的表示

三维空间变换

➤ 刚体运动变换:

• 旋转矩阵性质

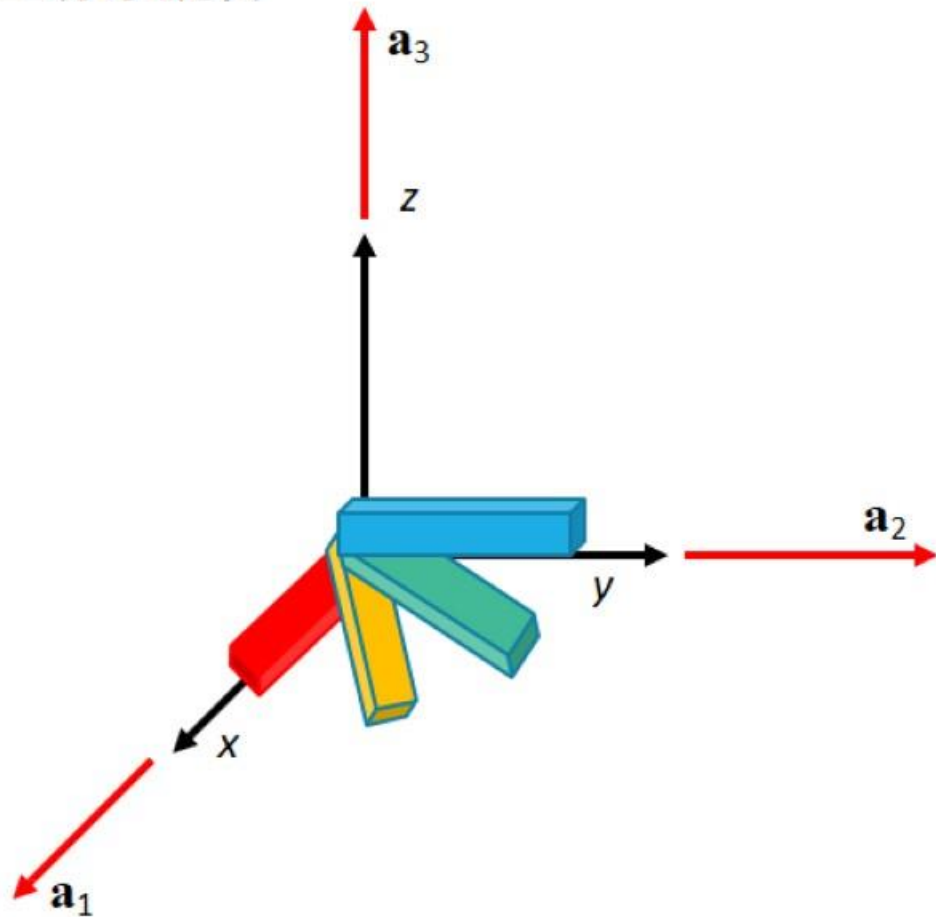
$$\mathbf{R}_B^A = \begin{bmatrix} \mathbf{b}_1^T \mathbf{a}_1 & \mathbf{b}_2^T \mathbf{a}_1 & \mathbf{b}_3^T \mathbf{a}_1 \\ \mathbf{b}_1^T \mathbf{a}_2 & \mathbf{b}_2^T \mathbf{a}_2 & \mathbf{b}_3^T \mathbf{a}_2 \\ \mathbf{b}_1^T \mathbf{a}_3 & \mathbf{b}_2^T \mathbf{a}_3 & \mathbf{b}_3^T \mathbf{a}_3 \end{bmatrix} \quad \mathbf{b}_i^T \mathbf{b}_i = 1 \quad \mathbf{b}_i^T \mathbf{b}_j = 0 \text{ with } i \neq j$$

- 所以旋转矩阵是正交矩阵
- $\mathbf{R}\mathbf{R}^T = \mathbf{R}^T\mathbf{R} = \mathbf{I}$
- 6个约束, 3个独立变量
- 特殊正交阵
- 如果是右手系, 行列式为+1
- **乘法封闭** 两个旋转矩阵相乘仍是旋转矩阵
- **旋转矩阵的逆**也是旋转矩阵 旋转矩阵的逆是自身的**转置**

三维空间变换

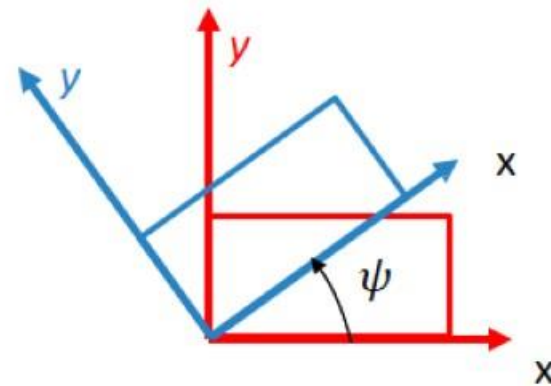
➤ 刚体运动变换:

- 坐标系旋转



- 绕 z 轴旋转

$$R_{z,\psi} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

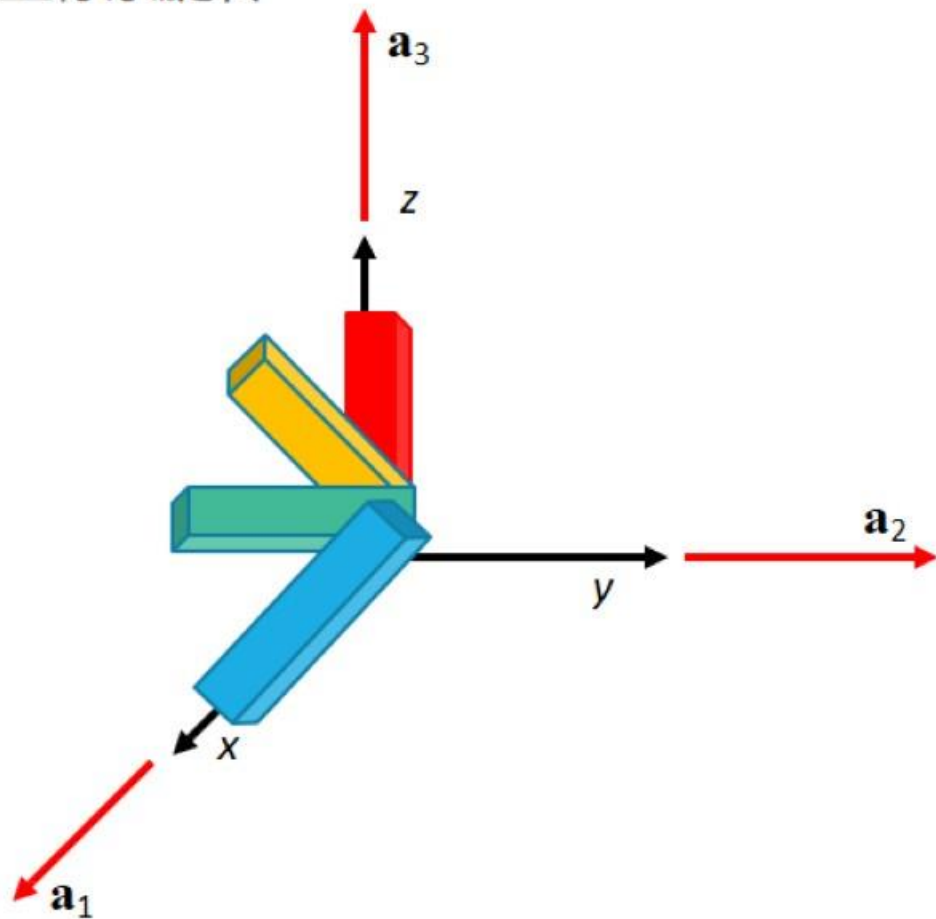


- R_B^A 表示在A坐标系下B坐标系的旋转

三维空间变换

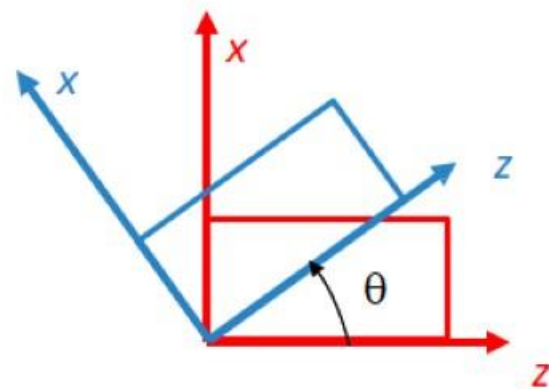
➤ 刚体运动变换:

- 坐标系旋转



- 绕y轴旋转

$$R_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

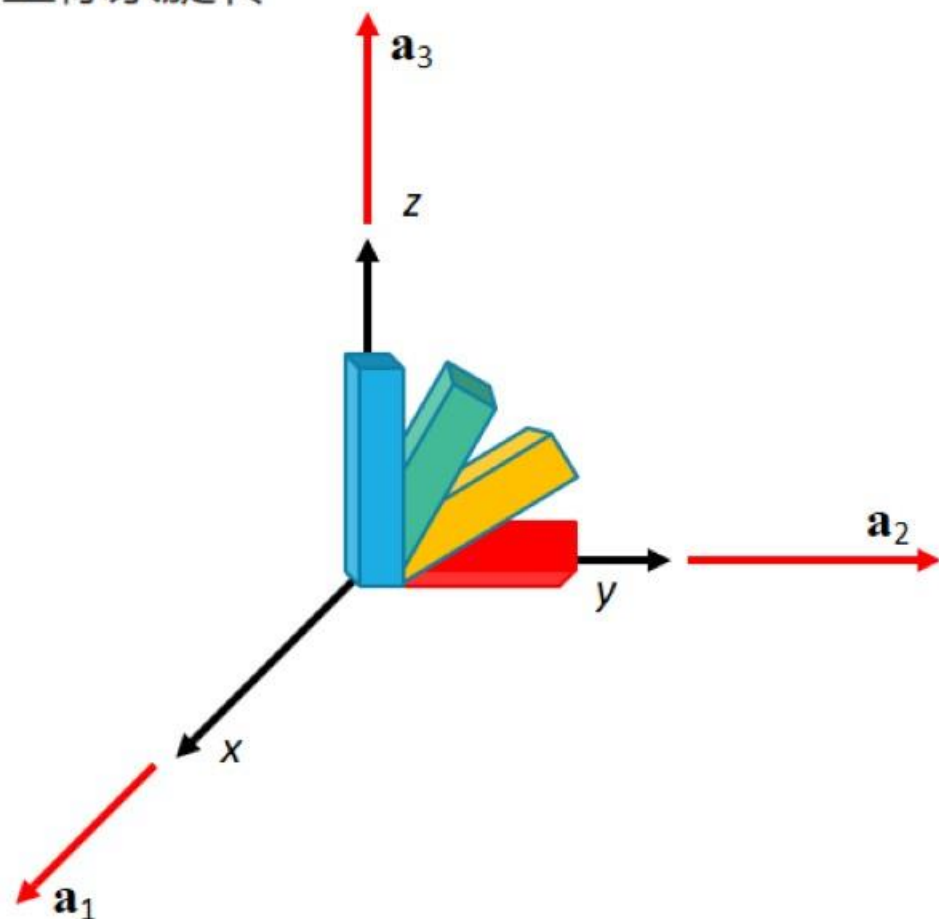


- R_B^A 表示在A坐标系下B坐标系的旋转

三维空间变换

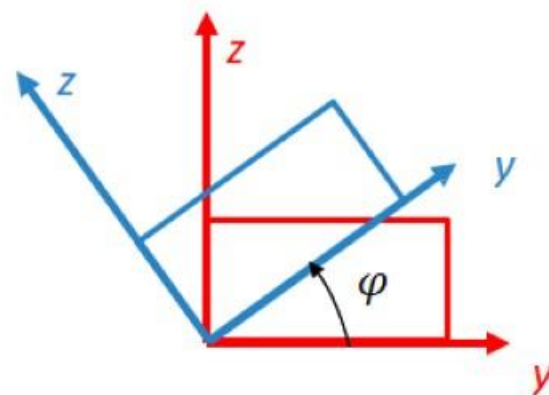
➤ 刚体运动变换:

- 坐标系旋转



- 绕x轴旋转

$$R_{x,\varphi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\varphi) & \cos(\varphi) \end{bmatrix}$$

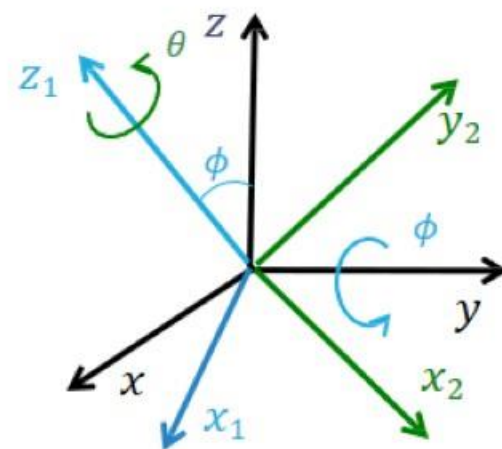
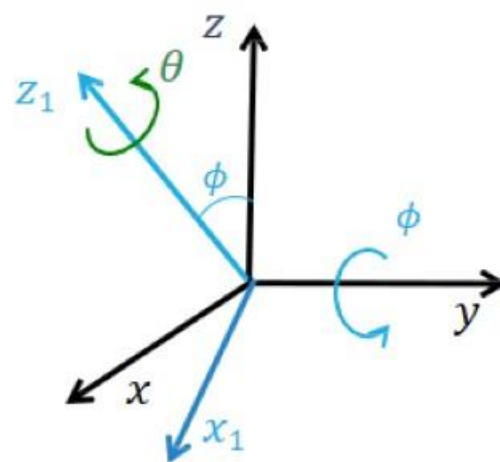
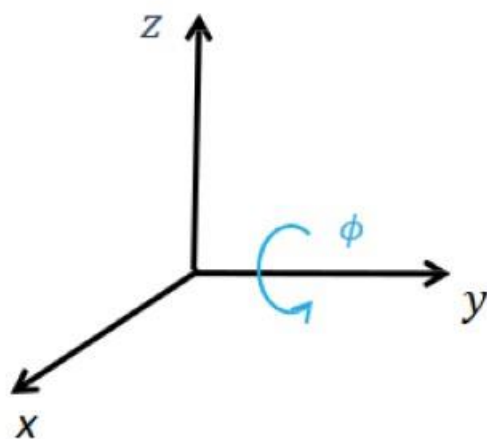


- R_B^A 表示在A坐标系下B坐标系的旋转

三维空间变换

➤ 刚体运动变换:

- 复合旋转



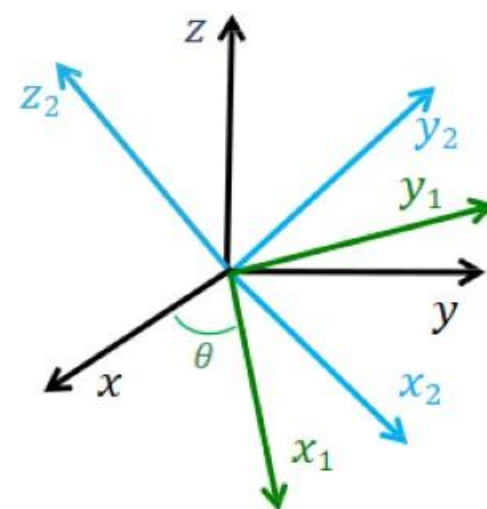
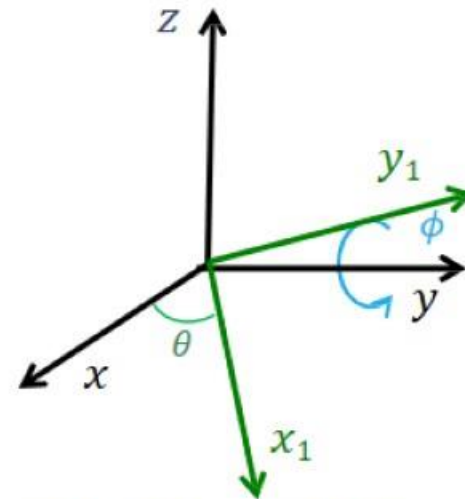
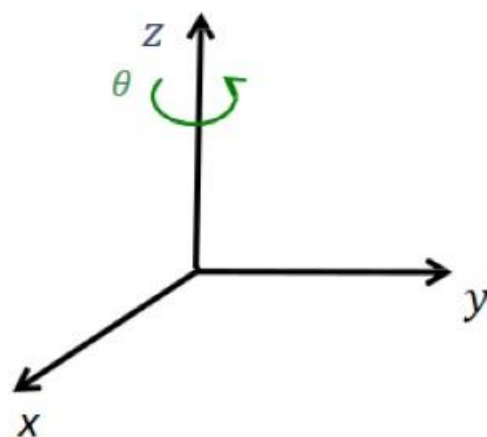
$$\mathbf{R}_2^0 = \mathbf{R}_1^0 \mathbf{R}_2^1 = R_{y,\phi} R_{z,\theta} \text{ 复合旋转在当前坐标系}$$

Post-multiply successive transformations about intermediate frames

• 两种旋转等价

固定坐标系下的复合旋转可以通过次序相反的中间坐标系旋转获得

$$R_{x_c,\phi} R_{y_c,\phi} R_{z_c,\theta} = R_{z_f,\theta} R_{y_f,\phi} R_{x_f,\phi}$$



$$\mathbf{R}_2^0 = \mathbf{R}_2^1 \mathbf{R}_1^0 = R_{z,\theta} R_{y,\phi} \text{ 复合旋转在固定坐标系}$$

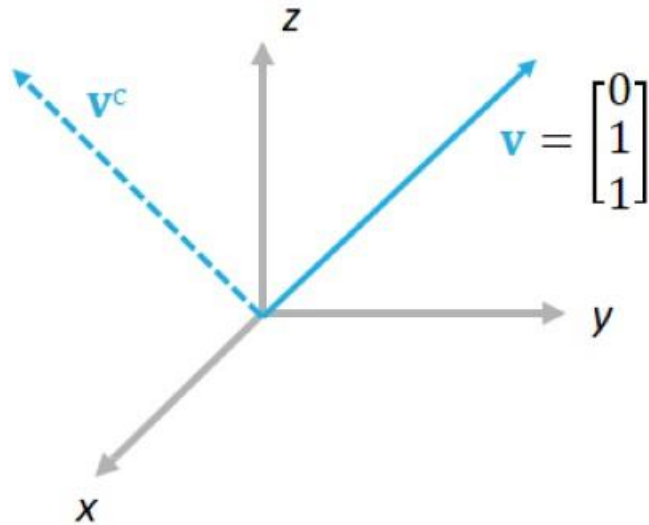
Pre-multiply successive transformations about fixed frames

三维空间变换

➤ 刚体运动变换:

• 向量旋转

将向量 $\mathbf{v}$ 绕x轴旋转 $\theta = \frac{\pi}{2}$



$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{v}' = R \mathbf{v}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$R_{x_c,\phi} R_{y_c,\phi} R_{z_c,\theta} = R_{z_f,\theta} R_{y_f,\phi} R_{x_f,\phi}$$

三维空间变换

➤ 刚体运动变换:

- 齐次变换 (旋转+平移)

齐次变换是用矩阵来表示刚体变换, 包括旋转和平移

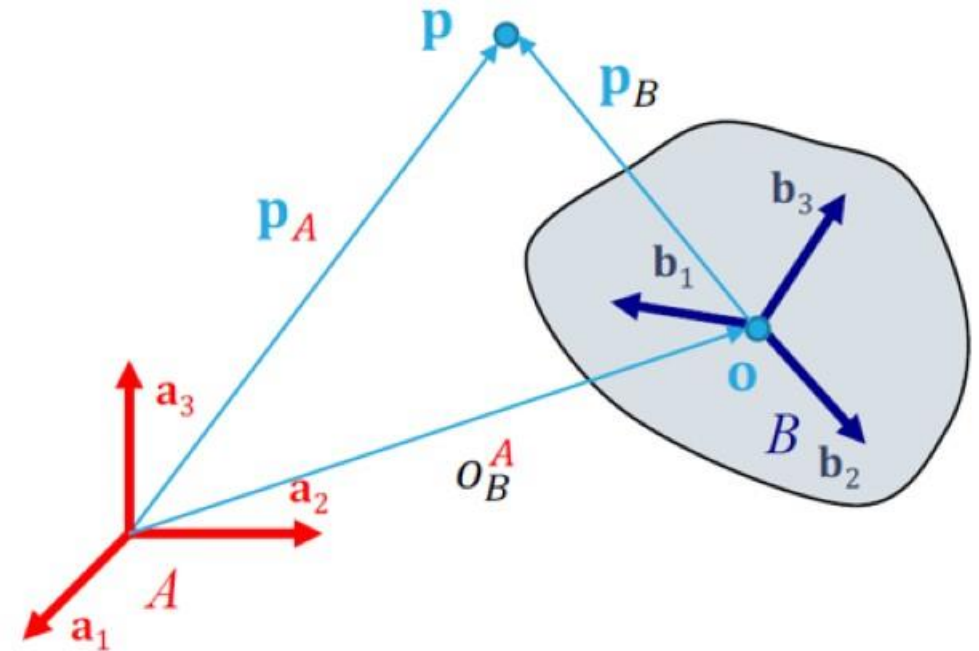
$$H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}$$

R 是 3×3 旋转矩阵, d 是 3×1 平移向量

向量的齐次表示为

$$P = \begin{bmatrix} p \\ 1 \end{bmatrix}$$

p 是 3×1 的向量



$$p_A = R_B^A p_B + o_B^A$$

$$P_A = H_B^A P_B = \begin{bmatrix} R_B^A & o_B^A \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_B \\ 1 \end{bmatrix} = \begin{bmatrix} R_B^A p_B + o_B^A \\ 1 \end{bmatrix} = \begin{bmatrix} p_A \\ 1 \end{bmatrix}$$

三维空间变换

➤ 刚体运动变换:

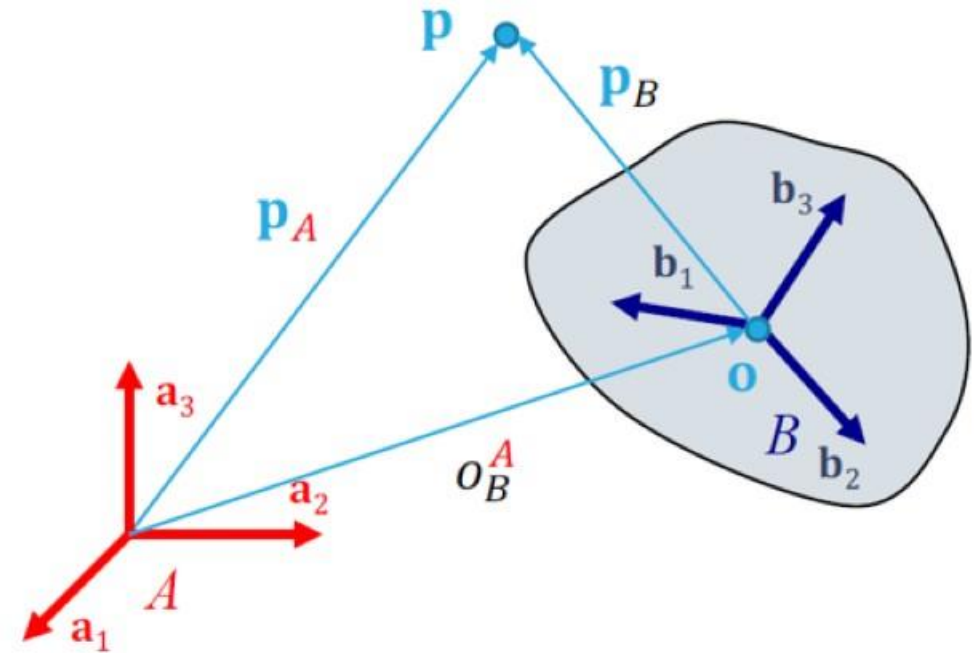
• 复合变换

$$H = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ 0 & 1 \end{bmatrix}$$

注意:

$$\begin{bmatrix} \mathbf{I} & \mathbf{d} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ 0 & 1 \end{bmatrix}$$

1. Can be defined with respect to the current frame
 - **Post**-multiply successive transformations about the intermediate frames
2. Can be defined with respect to the fixed frame
 - **Pre**-multiply successive transformations about the fixed frame



三维空间变换

➤ 刚体运动变换:

- 齐次变换的逆

$$\mathbf{H} = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ \mathbf{0} & 1 \end{bmatrix}$$

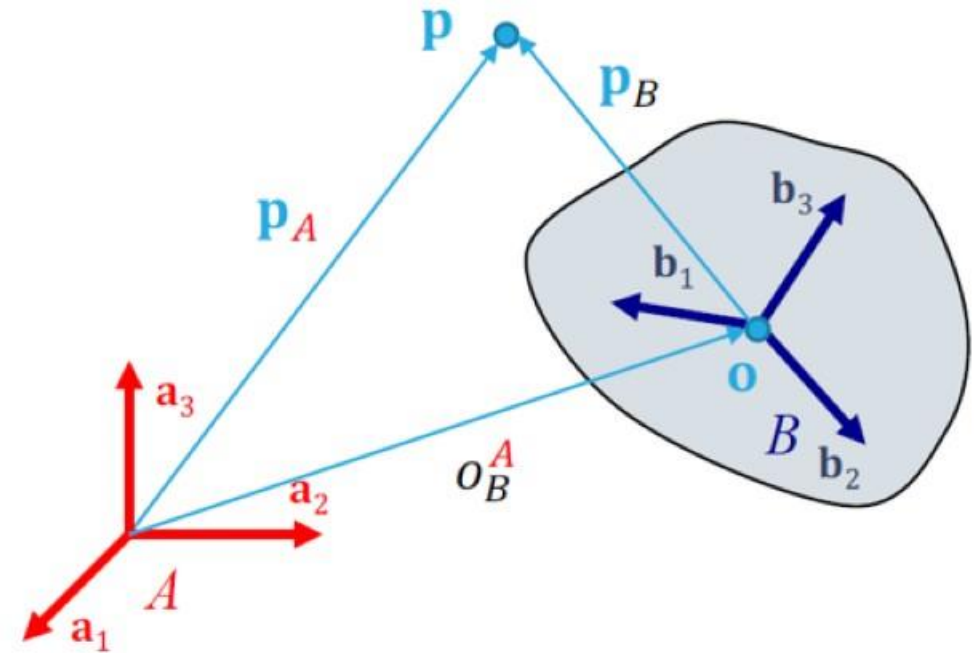
齐次变换的逆:

$$\mathbf{H}^{-1} = \begin{bmatrix} \mathbf{R}^T & -\mathbf{R}^T \mathbf{d} \\ \mathbf{0} & 1 \end{bmatrix} \neq \mathbf{H}^T$$

$$\begin{bmatrix} \mathbf{R}^T & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{d} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}^T & -\mathbf{R}^T \mathbf{d} \\ \mathbf{0} & 1 \end{bmatrix}$$

Interpretation:

1. Translation of $-\mathbf{d}$ followed by rotation $\mathbf{R}^T$ in the original frame
2. Rotation $\mathbf{R}^T$ followed by translation of $-\mathbf{d}$ in the current frame



三维空间变换

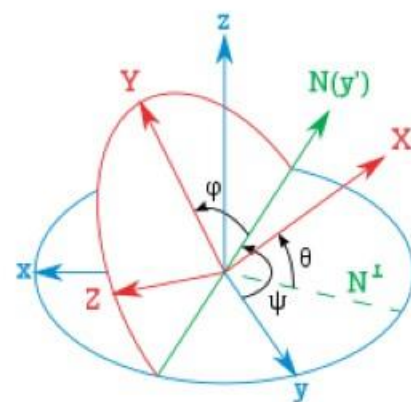
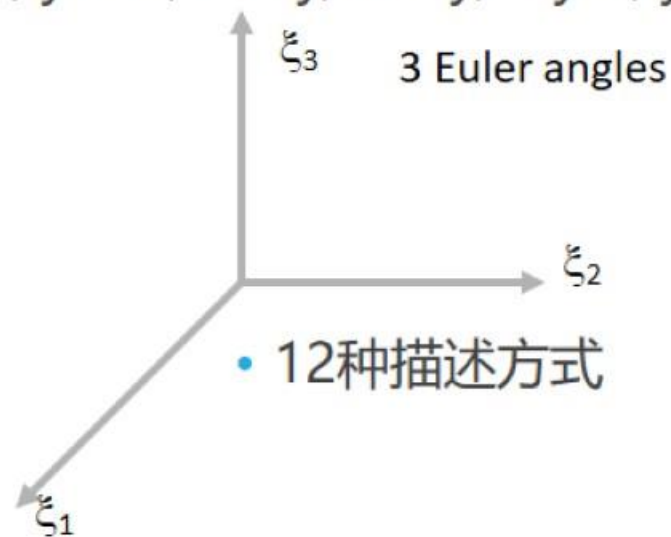
➤ 刚体运动变换:

• 欧拉角 (Euler Angle)

任何旋转都可以用三个**线性无关轴**的连续旋转来描述

- 两次非连续旋转中使用相同的轴
- 经典欧拉角(Proper Euler angles) (Z-X-Z, X-Y-X, Y-Z-Y, Z-Y-Z, X-Z-X, Y-X-Y)
- 不同的轴
- 泰特-布莱恩角(Tait-Bryan angles) (X-Y-Z, Y-Z-X, Z-X-Y, X-Z-Y, Z-Y-X, Y-X-Z)

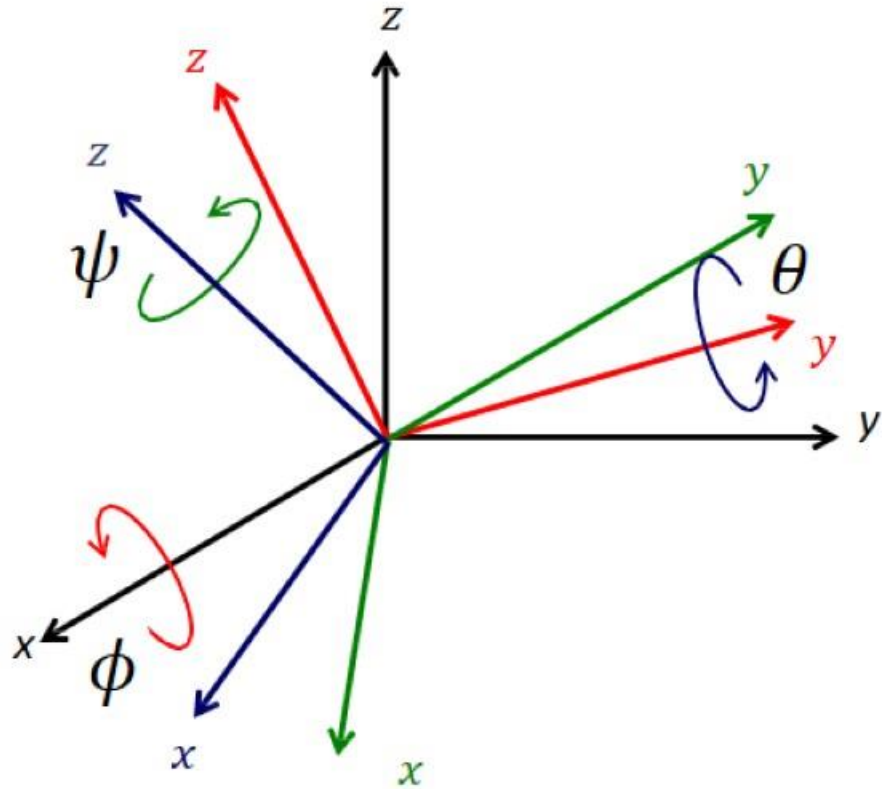
3×3
旋转矩
阵



三维空间变换

➤ 刚体运动变换:

• XYZ欧拉角

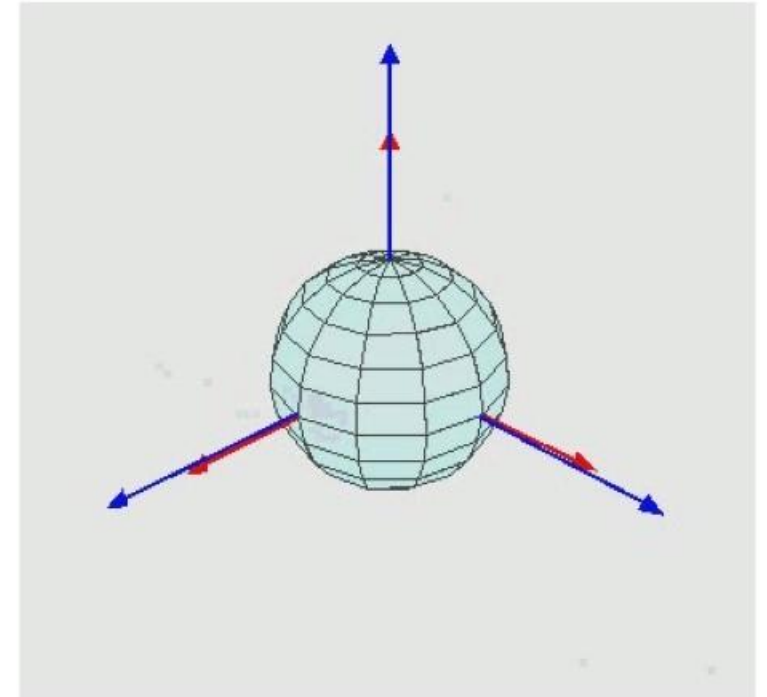


按机体当前坐标轴的顺序旋转

- $R_{x,\phi}$
- $R_{y,\theta}$
- $R_{z,\psi}$

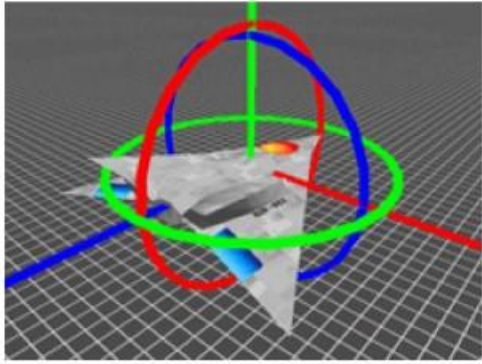
$$R = R_{x,\phi} R_{y,\theta} R_{z,\psi}$$

Roll, pitch, yaw

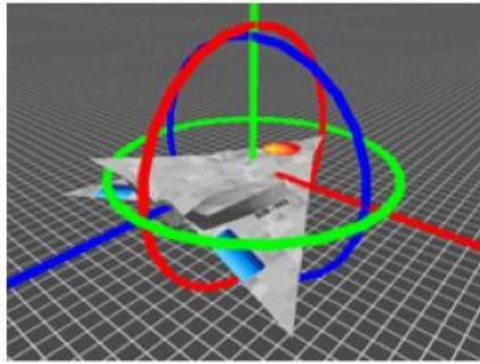


三维空间变换

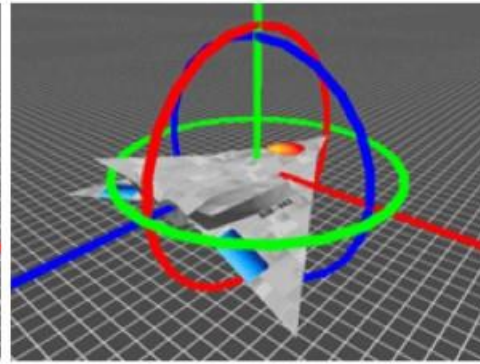
- 刚体运动变换:
 - XYZ欧拉角



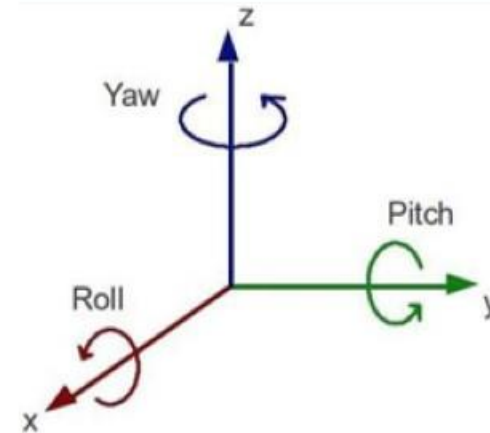
Roll, 桶滚



Yaw, 偏航

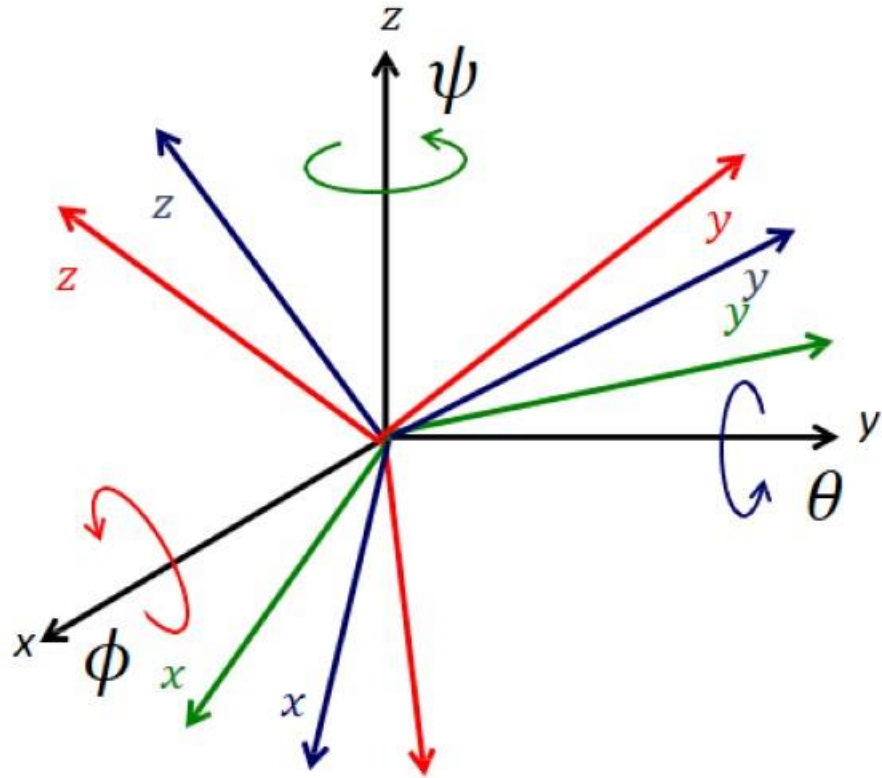


Pitch, 俯仰



三维空间变换

- 刚体运动变换:
 - ZYX欧拉角

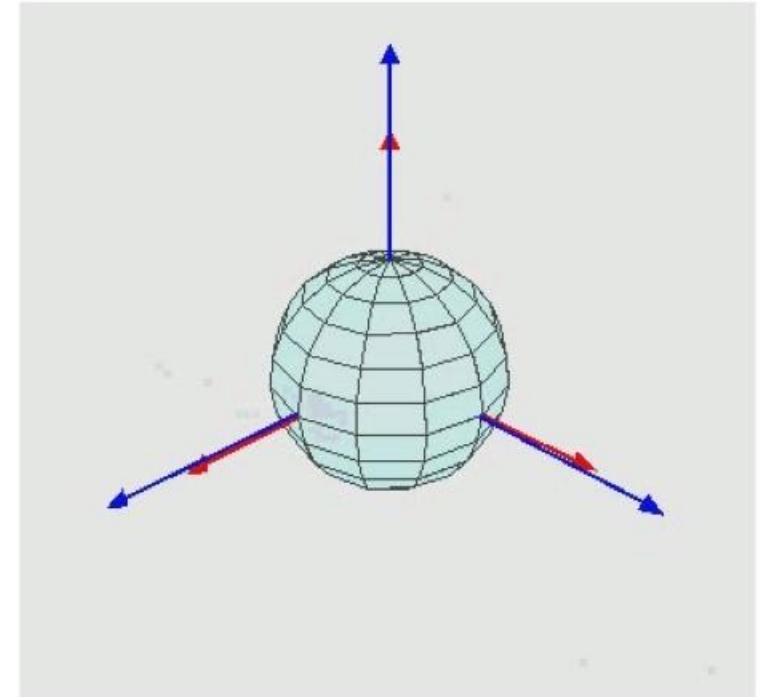


按固定坐标轴的顺序旋转

- $R_{z,\psi}$
- $R_{y,\theta}$
- $R_{x,\phi}$

$$R = R_{x,\phi} R_{y,\theta} R_{z,\psi}$$

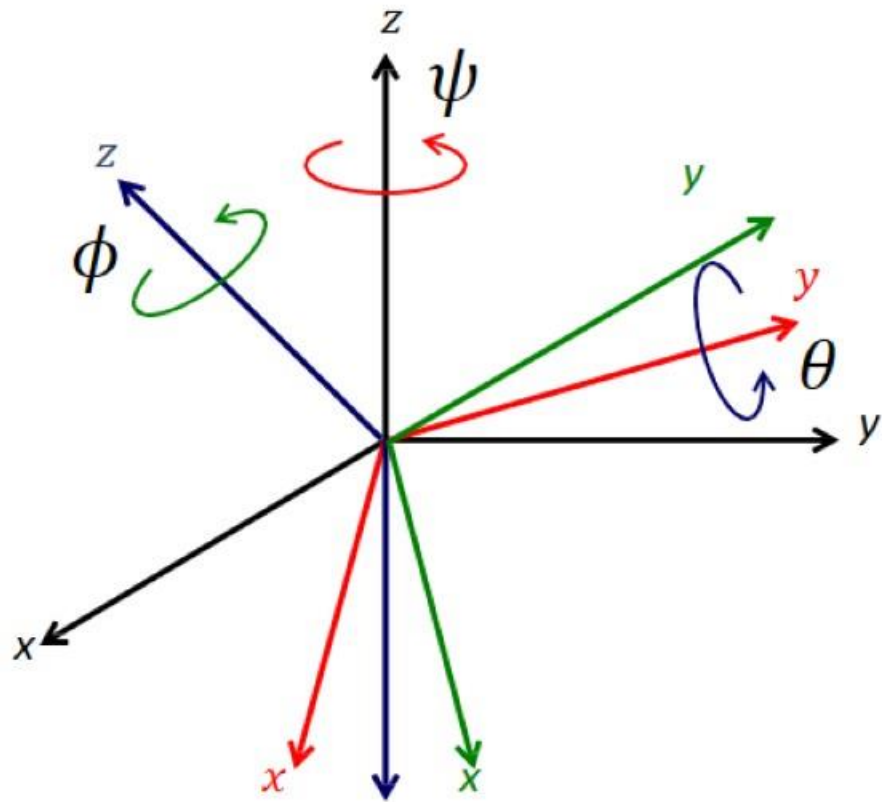
Roll, pitch, yaw



三维空间变换

➤ 刚体运动变换:

• ZYZ欧拉角



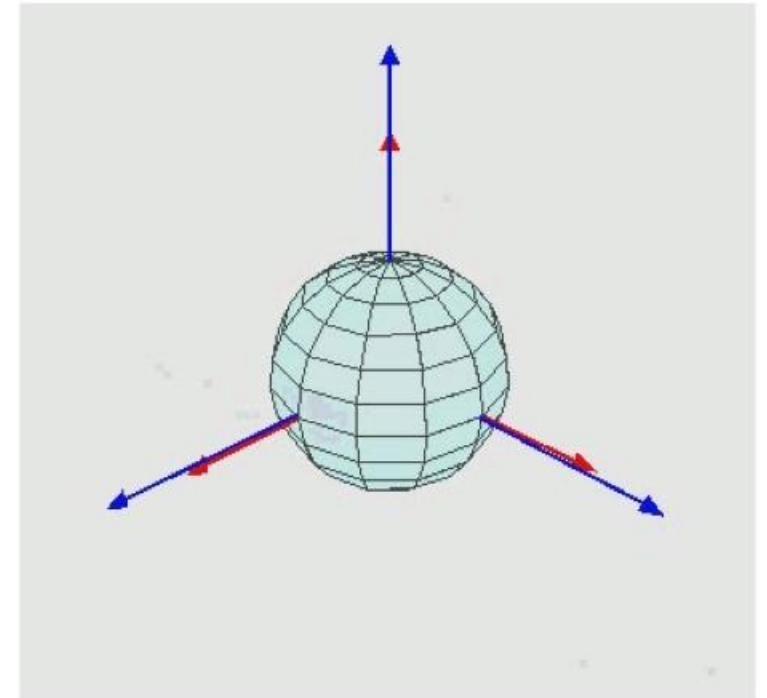
按机体当前坐标轴的顺序旋转

- $R_{z,\phi}$

- $R_{y,\theta}$

- $R_{z,\psi}$

$$R = R_{z,\phi} R_{y,\theta} R_{z,\psi}$$



三维空间变换

按机体当前坐标轴的顺序旋转

➤ 刚体运动变换:

• 欧拉角的确定

$$R = R_{z,\phi} R_{y,\theta} R_{z,\psi}$$

$$= \begin{bmatrix} \cos \phi \cos \theta \cos \psi - \sin \phi \sin \psi & -\cos \phi \cos \theta \sin \psi - \sin \phi \cos \psi & \cos \phi \sin \theta \\ \sin \phi \cos \theta \cos \psi + \cos \phi \sin \psi & -\sin \phi \cos \theta \sin \psi + \cos \phi \cos \psi & \sin \phi \sin \theta \\ -\sin \theta \cos \psi & \sin \theta \sin \psi & \cos \theta \end{bmatrix} =$$

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$\phi = \text{atan2}(R_{23}, R_{13})$$

$$\theta = \text{atan2}\left(\sqrt{R_{13}^2 + R_{23}^2}, R_{33}\right) \text{ with } 0 < \theta < \pi$$

$$\psi = \text{atan2}(R_{32}, -R_{31})$$

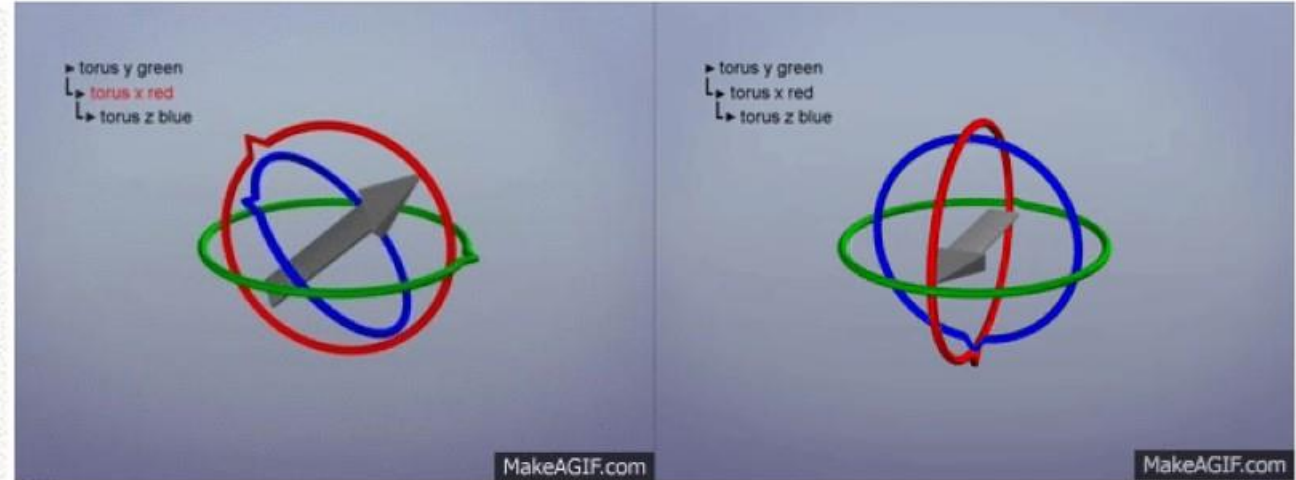
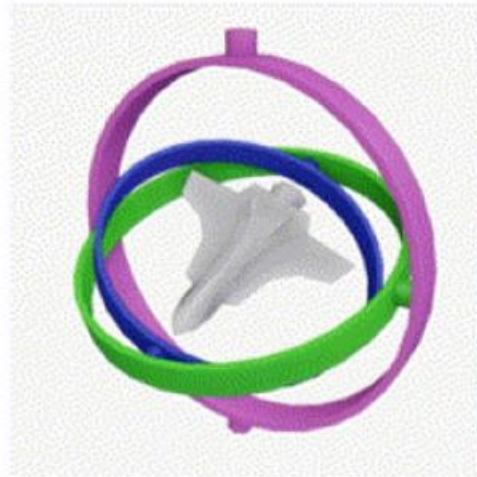
$$\text{atan2}(y, x) = \begin{cases} \arctan\left(\frac{y}{x}\right) & x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & x < 0 \text{ and } y \leq 0 \\ -\frac{\pi}{2} & x = 0 \text{ and } y < 0 \\ \frac{\pi}{2} & x = 0 \text{ and } y > 0 \\ \text{undefined} & x = 0 \text{ and } y = 0 \end{cases}$$

三维空间变换

➤ 刚体运动变换:

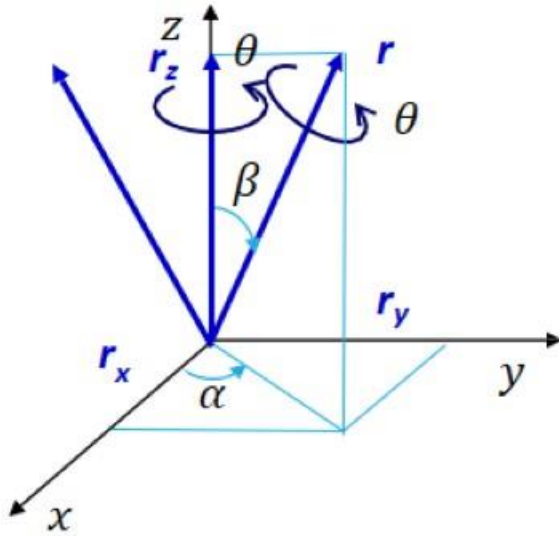
- 欧拉角与万向节死锁 (Gimbal Lock)

- **优点:**
- 容易理解
- **缺点:**
- 多种描述转换
- 描述不唯一
- 奇异值-万向节死锁



三维空间变换

- 刚体运动变换:
 - 旋转矩阵的轴角



$$\|r\| = 1$$

$$\mathbf{R}(\theta, r) = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_z(\theta) \mathbf{R}_y(-\beta) \mathbf{R}_z(-\alpha)$$

$$\sin \alpha = \frac{r_y}{\sqrt{r_x^2 + r_y^2}} \quad \cos \alpha = \frac{r_x}{\sqrt{r_x^2 + r_y^2}}$$

$$\sin \beta = \frac{r_z}{\sqrt{r_x^2 + r_y^2 + r_z^2}} \quad \cos \beta = \frac{r_z}{\sqrt{r_x^2 + r_y^2 + r_z^2}}$$

- 唯一性?

$$\mathbf{R}(\theta, r) = \mathbf{R}(-\theta, -r)$$

- 最小描述? :

$$\|r\| = \theta$$

三维空间变换

➤ 刚体运动变换:

- 反对称矩阵 (Skew Symmetric Matrices)

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \\ &= \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}\end{aligned}$$

- 反对称矩阵性质:

$$\mathbf{A}^T = -\mathbf{A}$$

- 符号标记:

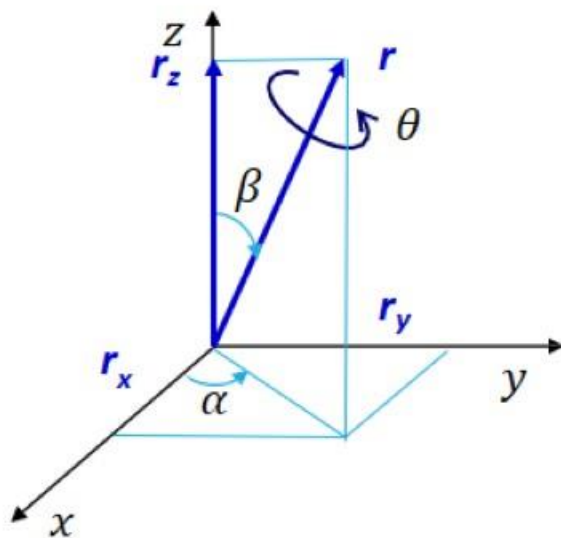
- $\hat{\mathbf{a}}$
- $\mathbf{S}(\mathbf{a})$
- $\mathbf{a}^\wedge$

可以从这种类型的矩阵中提取出三维向量 $\mathbf{a}$

三维空间变换

➤ 刚体运动变换:

- 旋转矩阵的轴角



$$\mathbf{R}(\theta, \mathbf{r}) = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_z(\theta) \mathbf{R}_y(-\beta) \mathbf{R}_z(-\alpha)$$

$$\sin \alpha = \frac{r_y}{\sqrt{r_x^2 + r_y^2}} \quad \cos \alpha = \frac{r_x}{\sqrt{r_x^2 + r_y^2}}$$

$$\sin \beta = \frac{\sqrt{r_x^2 + r_y^2}}{r} \quad \cos \beta = \frac{r_z}{r}$$

$$\mathbf{R}_{\mathbf{r}, \theta} = \mathbf{I} \cos \theta + \mathbf{r} \mathbf{r}^T (1 - \cos \theta) + \hat{\mathbf{r}} \sin \theta$$

$$\cos \theta = \frac{(\text{tr } \mathbf{R}) - 1}{2}$$

$$\hat{\mathbf{r}} = \frac{1}{2 \sin \theta} (\mathbf{R} - \mathbf{R}^T) \quad \mathbf{R} \mathbf{r} = \mathbf{r}$$

- 轴角表示转换到旋转矩阵是**多对一**
- 在 $[0, \pi]$ 内为**一对一**, 除了:

$$\theta = 0$$

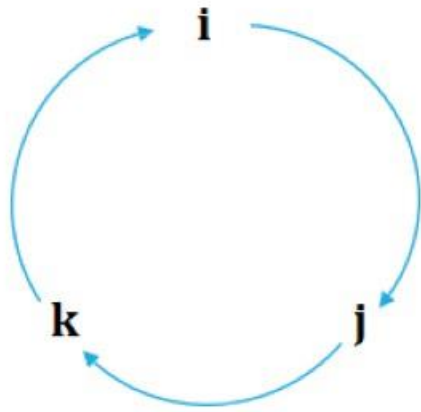
$$\theta = \pi$$

$\mathbf{r}$ 没有定义 $\mathbf{r}$ 不是唯一的, $\mathbf{r} = -\mathbf{r}$

三维空间变换

➤ 刚体运动变换:

- 四元数 (Quaternions)



- 复数的推广 $(1, \mathbf{i})$

基 $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$

$$\mathbf{ij} = \mathbf{k} \quad \mathbf{ji} = -\mathbf{k}$$

$$\mathbf{jk} = \mathbf{i} \quad \mathbf{kj} = -\mathbf{i}$$

$$\mathbf{ki} = \mathbf{j} \quad \mathbf{ik} = -\mathbf{j}$$

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$$

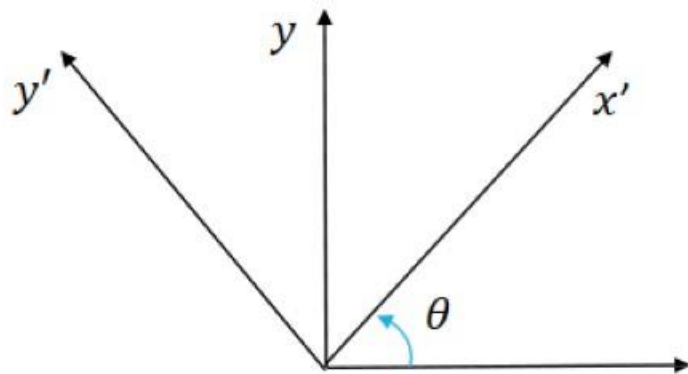
$$\mathbf{q} = q_w + \mathbf{q}_v$$

$$\mathbf{q} = q_w + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} = \begin{bmatrix} q_w \\ \mathbf{q}_v \end{bmatrix}$$

三维空间变换

➤ 刚体运动变换:

- 四元数 (Quaternions)



$$\mathbf{R} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos(\theta) + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \sin(\theta)$$

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{i} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{i}^2 = -\mathbf{I}$$

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \mathbf{I}a + \mathbf{i}b$$

- 这种类型的任何2x2矩阵都可以转换为复数
- 所有的复数都可以表示为这种类型的2x2矩阵
- 一个复数的平方模是这个矩阵的行列式
- $\|z_1 z_2\| = \|z_1\| \|z_2\|$ 来自于 $\det(A_1 A_2) = \det(A_1) \det(A_2)$
- 复数的逆相当于矩阵的逆

三维空间变换

➤ 刚体运动变换:

• 四元数 (Quaternions)

可以用同样的方法把 a, b, c, d ，每个有序的四重 a, b, c, d 与一个矩阵联系起来

$$q = \begin{pmatrix} a + id & -b - ic \\ b - ic & a - id \end{pmatrix} = a\mathbf{I} + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$$

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \mathbf{i} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \quad \mathbf{k} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

两个矩阵的和和乘积是同一类型的另一个矩阵

四元数的性质可以从这个代数推导出来

- 四元数的平方模是矩阵的行列式
- 对四元数矩阵求逆得到四元数逆公式
- 两个矩阵的和是另一个相同类型的矩阵，对于乘积也是一样的

三维空间变换

➤ 刚体运动变换:

- 单位四元数

四元数与轴角的关系:

$$w = \cos \frac{\theta}{2}$$

$$\mathbf{q}_v = \sin \frac{\theta}{2} \mathbf{r}$$

旋转并不唯一:

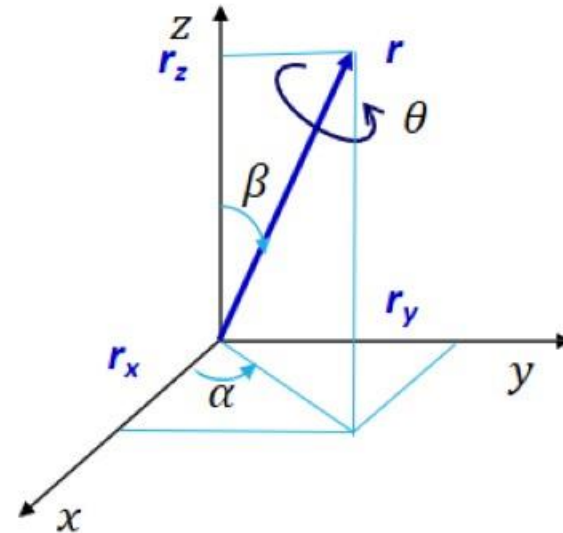
$$\mathbf{q}(\theta, \mathbf{r}) = \mathbf{q}(-\theta, -\mathbf{r})$$

求和可交换:

$$\mathbf{q}_1 + \mathbf{q}_2 = \begin{bmatrix} q_{w1} + q_{w2} \\ \mathbf{q}_{v1} + \mathbf{q}_{v2} \end{bmatrix} = \mathbf{q}_2 + \mathbf{q}_1$$

共轭:

$$\bar{\mathbf{q}} = \cos \frac{\theta}{2} - \sin \frac{\theta}{2} \mathbf{r}$$



三维空间变换

➤ 刚体运动变换:

• 四元数性质

逆:

$$\mathbf{q}^{-1} = \frac{\bar{\mathbf{q}}}{\|\mathbf{q}\|}$$

模:

$$\|\mathbf{q}\| = 1 = \sqrt{\bar{\mathbf{q}}\mathbf{q}} = \sqrt{\eta^2 + \epsilon_x^2 + \epsilon_y^2 + \epsilon_z^2}$$

乘积 (不可交换):

$$\mathbf{q}_1 * \mathbf{q}_2 = q_{w1}q_{w2} - \mathbf{q}_{v1}^T \mathbf{q}_{v2} + q_{w1}\mathbf{q}_{v2} + q_{w2}\mathbf{q}_{v1} + \mathbf{q}_{v1} \times \mathbf{q}_{v2} \neq \mathbf{q}_2 * \mathbf{q}_1$$

结合律:

$$(\mathbf{q}_1 * \mathbf{q}_2) * \mathbf{q}_3 = \mathbf{q}_1 * (\mathbf{q}_2 * \mathbf{q}_3)$$

分配律:

$$(\mathbf{q}_1 + \mathbf{q}_2) * \mathbf{q}_3 = \mathbf{q}_1 * \mathbf{q}_3 + \mathbf{q}_2 * \mathbf{q}_3$$

三维空间变换

➤ 刚体运动变换:

• 四元数性质

乘积:

$$\mathbf{q}_1 * \mathbf{q}_2 = q_{w1}q_{w2} - \mathbf{q}_{v1}^T \mathbf{q}_{v2} + q_{w1}\mathbf{q}_{v2} + q_{w2}\mathbf{q}_{v1} + \mathbf{q}_{v1} \times \mathbf{q}_{v2} \neq \mathbf{q}_2 * \mathbf{q}_1$$

$$q_1 = a + bi + cj + dk, q_2 = e + fi + gj + hk :$$

$$q_1 q_2 = (a + bi + cj + dk)(e + fi + gj + hk)$$

$$= ae + afi + agj + ahk +$$

$$bei + bfi^2 + bgij + bhik +$$

$$cej + cfji + cgj^2 + chjk +$$

$$dke + dfki + dgkj + dhk^2$$

$$= (ae - bf - cg - dh) +$$

$$(be + af - dg + ch)i +$$

$$(ce + df + ag - bh)j +$$

$$(de - cf + bg + ah)k$$

$$q_1 q_2 = \begin{bmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & -c & b & a \end{bmatrix} \begin{bmatrix} e \\ f \\ g \\ h \end{bmatrix}$$

$$\text{又有 } q_1 = [a, \mathbf{v}], \mathbf{v} = \begin{bmatrix} b \\ c \\ d \end{bmatrix}, q_2 = [e, \mathbf{u}], \mathbf{u} = \begin{bmatrix} f \\ g \\ h \end{bmatrix}$$

$$\mathbf{v} \cdot \mathbf{u} = bf + cg + dh, \mathbf{v} \times \mathbf{u} = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b & c & d \\ f & g & h \end{bmatrix} = (ch - dg)\mathbf{i} - (bh - df)\mathbf{j} + (bg - cf)\mathbf{k}$$

$$q_1 q_2 = [ae - \mathbf{v} \cdot \mathbf{u}, a\mathbf{u} + e\mathbf{v} + \mathbf{v} \times \mathbf{u}]$$

三维空间变换

➤ 刚体运动变换:

- 四元数表示旋转

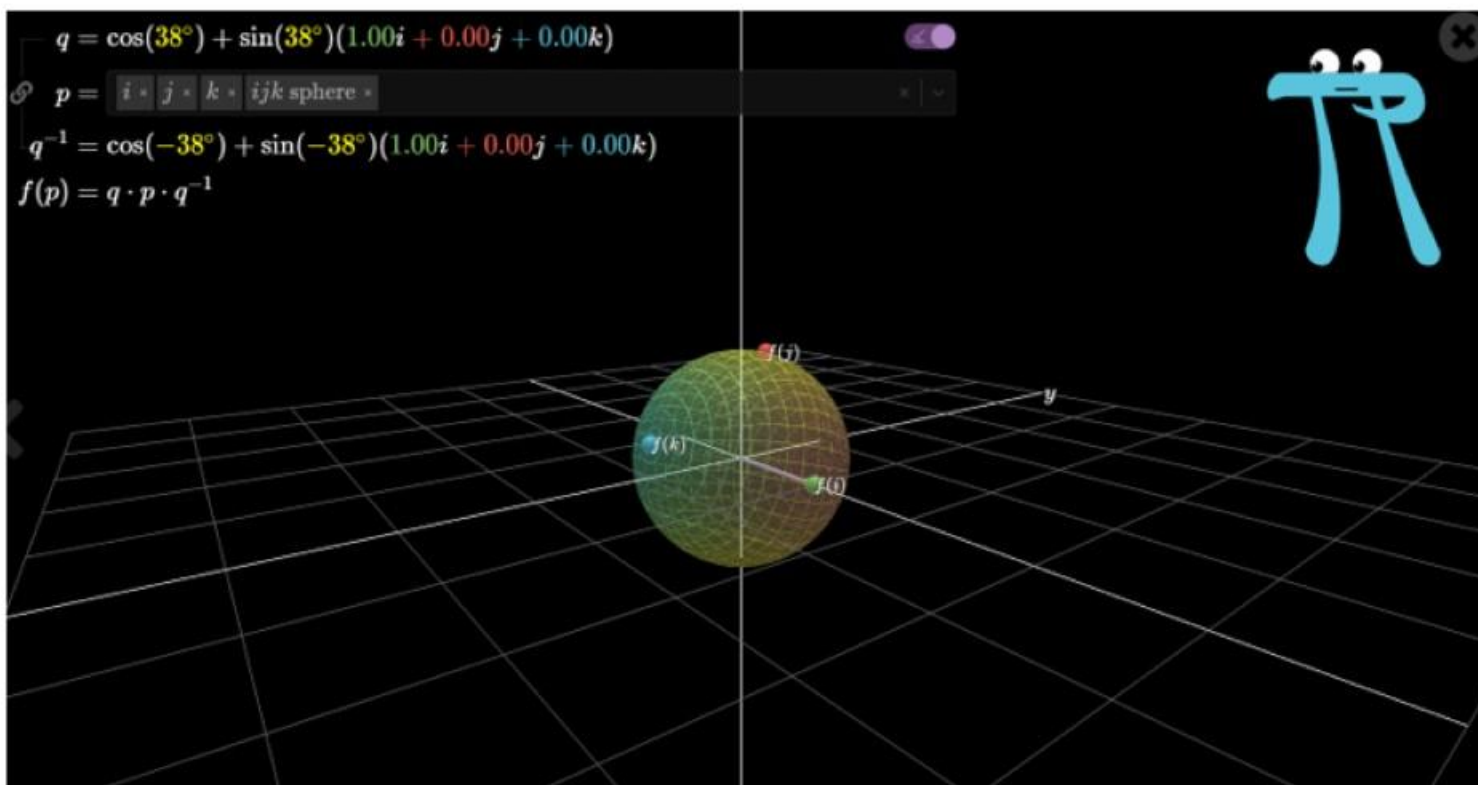
用四元数旋转一个向量:

$$\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \Rightarrow \mathbf{p}_a = \begin{bmatrix} 0 \\ p_x \\ p_y \\ p_z \end{bmatrix} \quad \text{虚四元数}$$

$$\mathbf{p}' = R\mathbf{p} \Leftrightarrow \mathbf{p}' = \mathbf{q} * \mathbf{p}_a * \mathbf{q}^{-1}$$

$$w = \cos \frac{\theta}{2}$$

$$\mathbf{q}_v = \sin \frac{\theta}{2} \mathbf{r}$$



三维空间变换

➤ 刚体运动变换:

• 四元数与旋转矩阵转换

$$q_1 = a + bi + cj + dk, q_2 = e + fi + gj + hk :$$

$$q_1 q_2 = (a + bi + cj + dk)(e + fi + gj + hk)$$

$$= ae + a fi + agj + ahk +$$

$$bei + b fi^2 + bgij + bhik +$$

$$cej + cfji + cgj^2 + chjk +$$

$$dke + dfki + dgkj + dhk^2$$

$$= (ae - bf - cg - dh) +$$

$$(be + af - dg + ch)i +$$

$$(ce + df + ag - bh)j +$$

$$(de - cf + bg + ah)k$$

$$\mathbf{p}' = R\mathbf{p} \Leftrightarrow \mathbf{p}' = \mathbf{q} * \mathbf{p}_a * \mathbf{q}^{-1}$$

$$q_1 q_2 = \begin{bmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & -c & b & a \end{bmatrix} \begin{bmatrix} e \\ f \\ g \\ h \end{bmatrix}$$

$$\mathbf{q}_1 * \mathbf{q}_2 = q_{w1}q_{w2} - \mathbf{q}_{v1}^T \mathbf{q}_{v2} + q_{w1}\mathbf{q}_{v2} + q_{w2}\mathbf{q}_{v1} + \mathbf{q}_{v1} \times \mathbf{q}_{v2} \neq \mathbf{q}_2 * \mathbf{q}_1$$

$$\mathbf{q}_1 * \mathbf{q}_2 = \left(Iq_w + \begin{bmatrix} 0 & -\mathbf{q}_{v1}^T \\ \mathbf{q}_{v1}^T & s(\mathbf{q}_{v1})^T \end{bmatrix} \right) \mathbf{q}_2$$

三维空间变换

➤ 刚体运动变换:

• 四元数与旋转矩阵转换

$$\mathbf{q} = [s, \mathbf{v}]^T$$

$$\mathbf{q}^+ = \begin{bmatrix} s & -\mathbf{v}^T \\ \mathbf{v} & s\mathbf{I} + \mathbf{v}^\wedge \end{bmatrix}, \quad \mathbf{q}^\oplus = \begin{bmatrix} s & -\mathbf{v}^T \\ \mathbf{v} & s\mathbf{I} - \mathbf{v}^\wedge \end{bmatrix}$$

$$\mathbf{q}_1^+ \mathbf{q}_2 = \begin{bmatrix} s_1 & -\mathbf{v}_1^T \\ \mathbf{v}_1 & s_1\mathbf{I} + \mathbf{v}_1^\wedge \end{bmatrix} \begin{bmatrix} s_2 \\ \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} -\mathbf{v}_1^T \mathbf{v}_2 + s_1 s_2 \\ s_1 \mathbf{v}_2 + s_2 \mathbf{v}_1 + \mathbf{v}_1^\wedge \mathbf{v}_2 \end{bmatrix} = \mathbf{q}_1 \mathbf{q}_2$$

$$\mathbf{q}_1 \mathbf{q}_2 = \mathbf{q}_1^+ \mathbf{q}_2 = \mathbf{q}_2^\oplus \mathbf{q}_1$$

三维空间变换

➤ 刚体运动变换:

• 四元数与旋转矩阵转换 $q_1 q_2 = q_1^+ q_2 = q_2^\oplus q_1$

$$\begin{aligned} p' &= qpq^{-1} = q^+ p^+ q^{-1} \\ &= q^+ q^{-1\oplus} p. \end{aligned}$$

$$q^+(q^{-1})^\oplus = \begin{bmatrix} s & -v^T \\ v & sI + v^\wedge \end{bmatrix} \begin{bmatrix} s & v^T \\ -v & sI + v^\wedge \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0^T & vv^T + s^2 I + 2sv^\wedge + (v^\wedge)^2 \end{bmatrix}$$

$$R = vv^T + s^2 I + 2sv^\wedge + (v^\wedge)^2$$

$$\cos \theta = 2s^2 - 1 = 2 \cos^2 \frac{\theta}{2} - 1,$$

$$\theta = 2 \arccos s.$$

$$\begin{aligned} \text{tr}(R) &= \text{tr}(vv^T + 3s^2 + 2s \cdot 0 + \text{tr}((v^\wedge)^2)) \\ &= v_1^2 + v_2^2 + v_3^2 + 3s^2 - 2(v_1^2 + v_2^2 + v_3^2) \\ &= (1 - s^2) + 3s^2 - 2(1 - s^2) \\ &= 4s^2 - 1. \end{aligned}$$

$$\begin{cases} \theta = 2 \arccos q_0 \\ [n_x, n_y, n_z]^T = [q_1, q_2, q_3]^T / \sin \frac{\theta}{2} \end{cases}$$

三维空间变换

➤ 刚体运动变换:

• 四元数与旋转矩阵转换

$$R = vv^T + s^2 I + 2sv^\wedge + (v^\wedge)^2$$

$$q = (q_0, q_1, q_2, q_3)$$

$$R = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_1q_3 + 2q_0q_2 \\ 2q_1q_2 + 2q_0q_3 & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_2q_3 + 2q_0q_1 & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_1q_3 + 2q_0q_2 \\ 2q_1q_2 + 2q_0q_3 & 1 - 2q_1^2 - 2q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_2q_3 + 2q_0q_1 & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix}$$

$$q_0 = \frac{\sqrt{\text{tr}(R) + 1}}{2}, q_1 = \frac{m_{23} - m_{32}}{4q_0}, q_2 = \frac{m_{31} - m_{13}}{4q_0}, q_3 = \frac{m_{12} - m_{21}}{4q_0}$$

三维空间变换

➤ 刚体运动变换:

• 四元数与旋转矩阵转换

$$R = vv^T + s^2 I + 2sv^\wedge + (v^\wedge)^2$$

$$q = (q_0, q_1, q_2, q_3)$$

$$R = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_1q_3 + 2q_0q_2 \\ 2q_1q_2 + 2q_0q_3 & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_2q_3 + 2q_0q_1 & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_1q_3 + 2q_0q_2 \\ 2q_1q_2 + 2q_0q_3 & 1 - 2q_1^2 - 2q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_2q_3 + 2q_0q_1 & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix}$$

$$q_0 = \frac{\sqrt{\text{tr}(R) + 1}}{2}, q_1 = \frac{m_{23} - m_{32}}{4q_0}, q_2 = \frac{m_{31} - m_{13}}{4q_0}, q_3 = \frac{m_{12} - m_{21}}{4q_0}$$

三维空间变换

➤ 刚体运动变换:

• 四元数

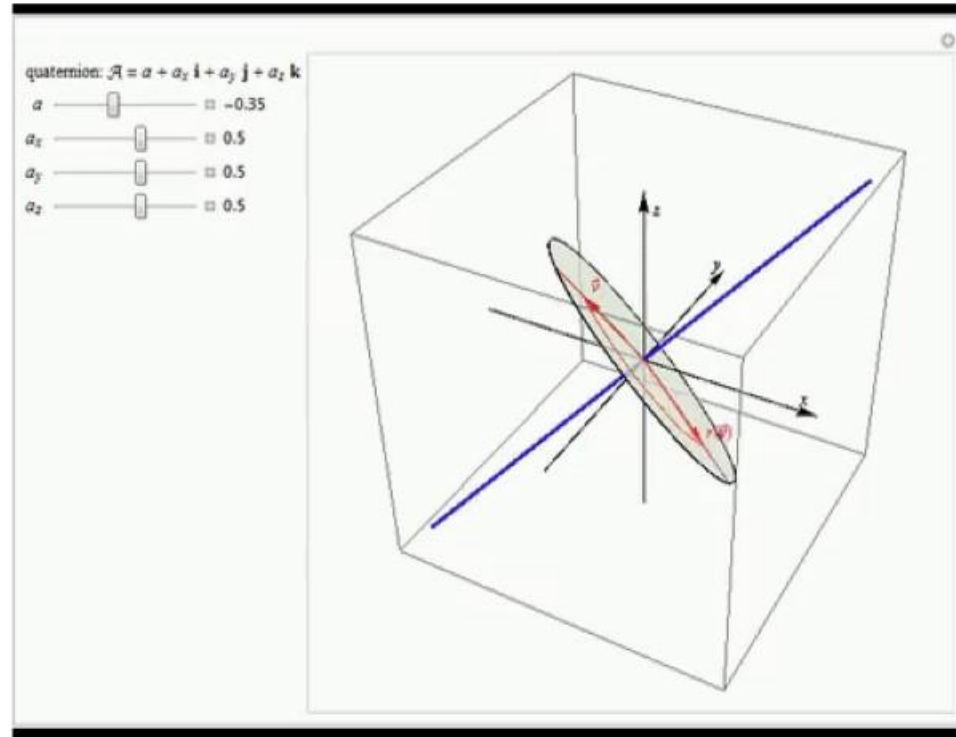
• 优点:

- 比旋转矩阵紧凑
- 容易与轴角表示转换
- 容易差值
- 数值计算稳定

• 缺点:

- 描述不唯一

$q, -q$ 是一样的



三维空间变换

- 刚体运动变换:
 - 总结

	Global	Unique	Minimal
旋转矩阵 Rotation Matrix	Yes	Yes	No-9 parameters
四元数 Quaternion	Yes	No	No-4 parameters
欧拉角 Euler Angles	No	No	Yes-3 parameters
轴角表示 Axis Angle	Yes	No	Yes-3 parameters

Global: 可以表示所有姿态

Unique: 每个姿态只有一种表示存在

Minimal: 能够用3个值来表示

三维空间变换

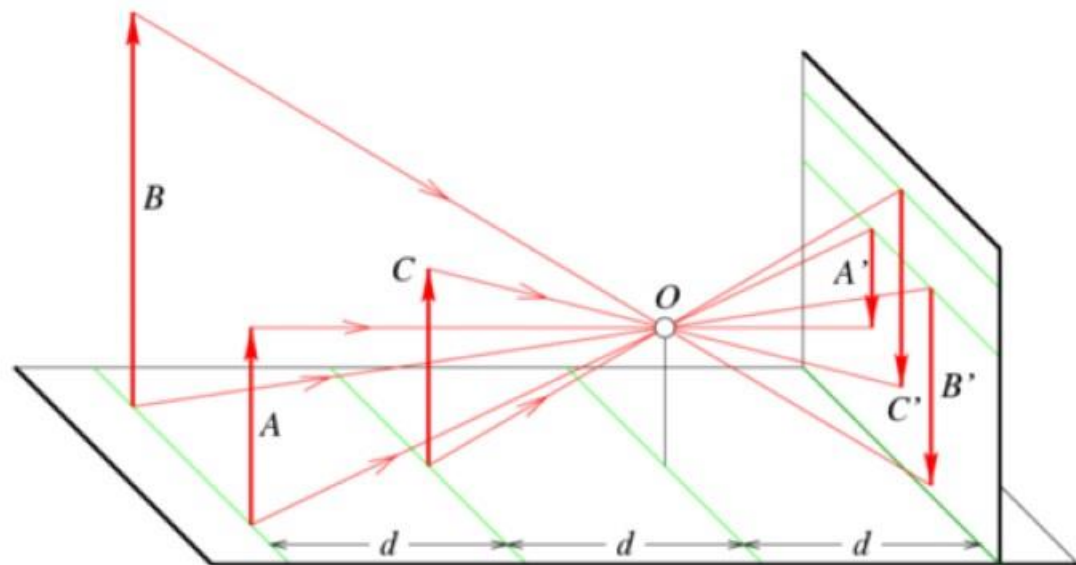
- 空间几何变换:
 - 投影变换



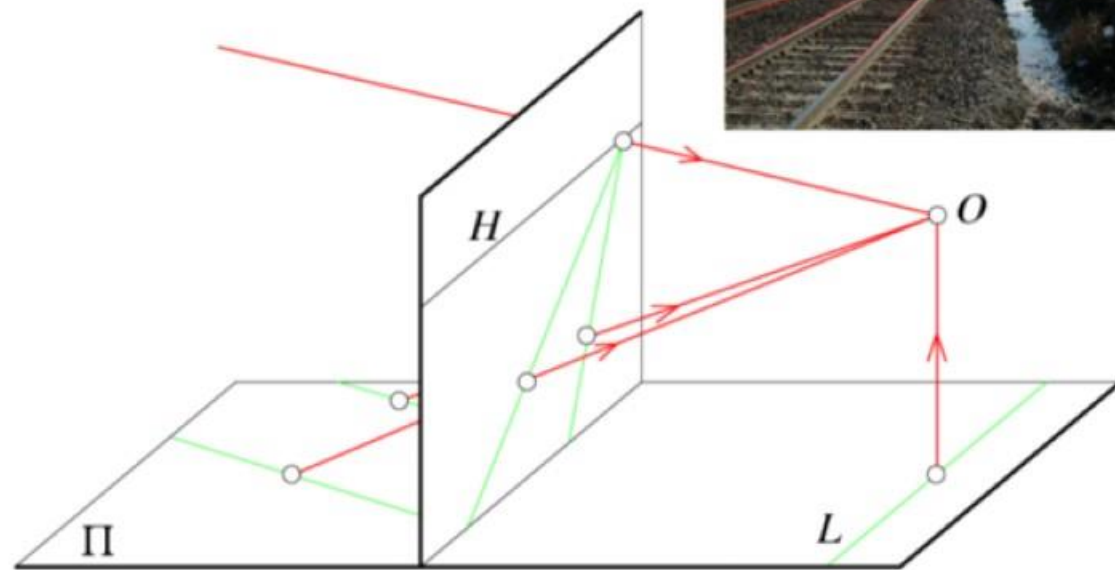
三维空间变换

➤ 空间几何变换:

- 投影变换



投影是等比例的



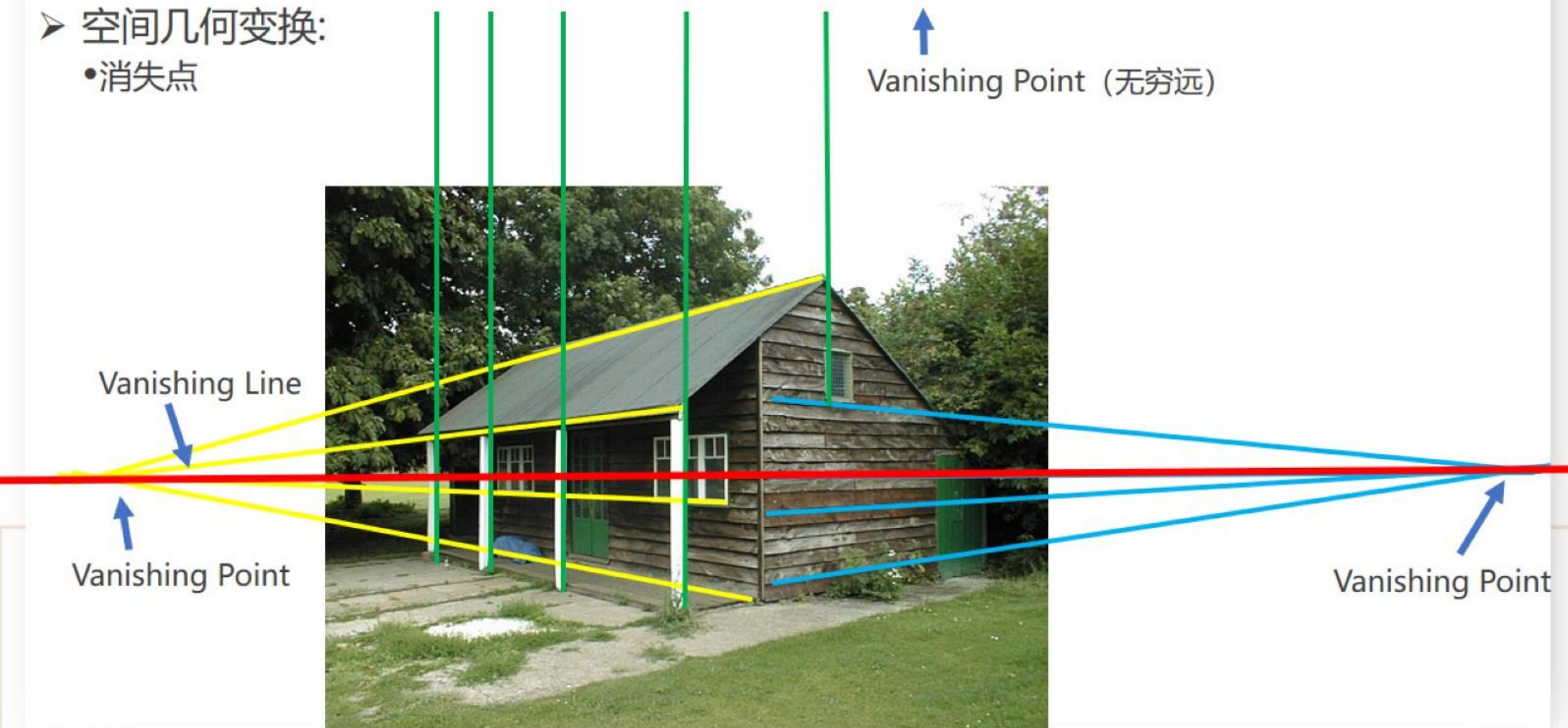
消失点/无穷远点 (Vanishing Point)



三维空间变换

➤ 空间几何变换:

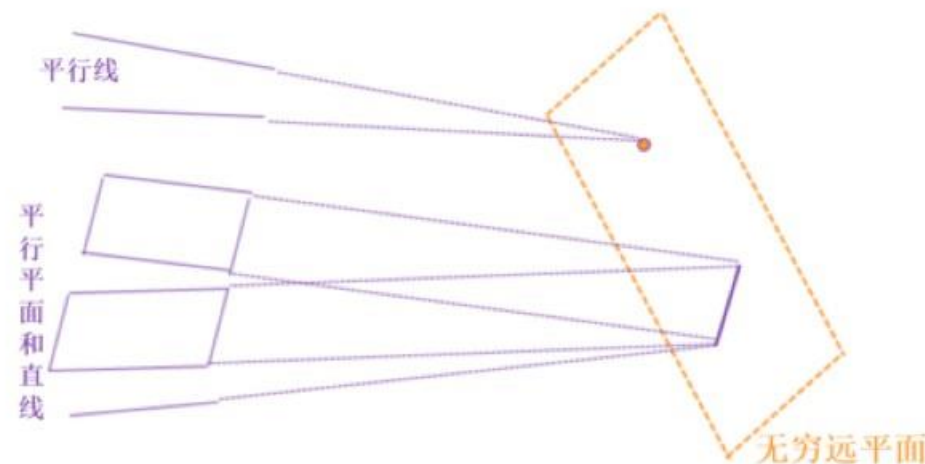
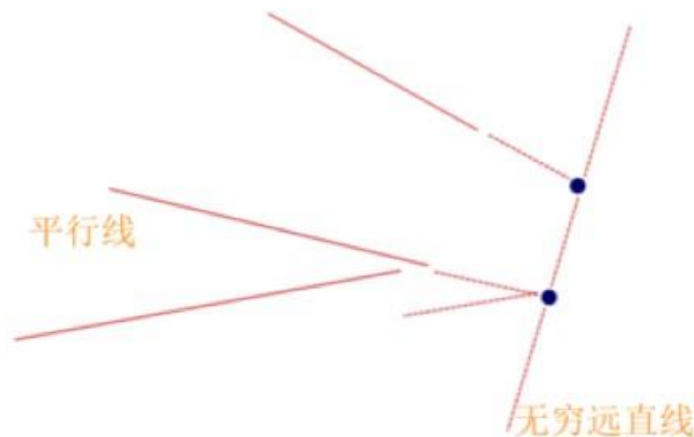
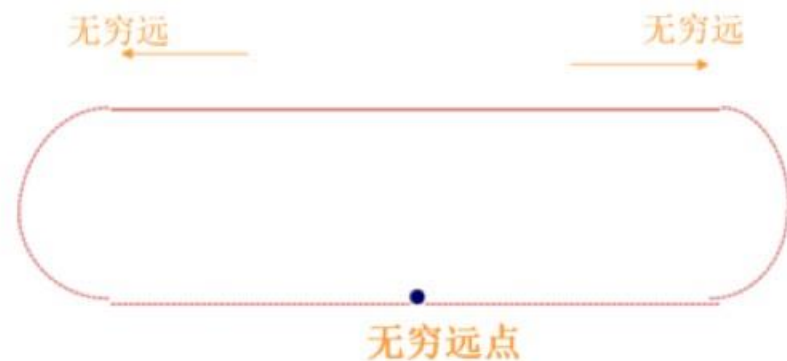
- 消失点



三维空间变换

➤ 空间几何变换:

- 无穷远元素



- **无穷远点**
平行线相交于无穷远点
- **无穷远直线**
平行平面相交于无穷远直线
- **无穷远平面**
所有无穷远点组成的平面即无穷远平面

- **n维射影空间**
- 1维射影空间:
射影直线, 欧式直线与无穷远点
- 2维射影空间
射影平面, 欧式平面与无穷远直线
- 3维射影空间
射影空间, 欧式空间与无穷远平面

三维空间变换

➤ 空间几何变换:

• 齐次坐标

在 n 维空间中, 建立欧氏坐标后, 每一个有限的点的坐标为 $(m_1, \dots, m_n)$, 对任意 $n+1$ 个数 $x_1, \dots, x_n, x_0$, 如果满足:

$$x_0 \neq 0, \frac{x_1}{x_0} = m_1, \dots, \frac{x_n}{x_0} = m_n.$$

则 $(x_1, \dots, x_n, x_0)$ 被叫作这个点的**齐次坐标**.
 $(m_1, \dots, m_n)$ 被称作**非齐次坐标**.

不全为0的数 $x_1, \dots, x_n$ 组成的坐标
 $(x_1, \dots, x_n, 0)$ 被称作**无穷远点的齐次坐标**.

$P_1(0,0)$	$P_1(0,0,x_3), (x_3 \neq 0)$	$P_1(0,0,1)$
$P_2(1,0)$	$P_2(\rho,0,\rho), (\rho \neq 0)$	$P_2(1,0,1)$

• 1维齐次坐标定义

	非齐次	关系	齐次坐标
有穷远点	x	$x = x_1/x_2$	$(x_1, x_2)(x_2 \neq 0)$
无穷远点	No	No	$(x_1, 0)(x_1 \neq 0)$

• 2维齐次坐标定义

	非齐次	关系	齐次坐标
有穷远点	(x, y)	$x = x_1/x_3,$ $y = x_2/x_3$	$(x_1, x_2, x_3)(x_3 \neq 0)$
无穷远点: 方向为 $\lambda = x_2/x_1$ 的无穷远点	No	No	$(x_1, x_2, 0)(x_1 \neq 0)$ $(\lambda = x_2/x_1)$
无穷远点: y轴上的无 穷远点	No	No	$(0, x_2, 0)(x_2 \neq 0)$

三维空间变换

➤ 空间几何变换:

- 齐次坐标

- 非齐次坐标变换为齐次坐标

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- 齐次坐标变换为非齐次坐标

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

➤ 为什么要用齐次坐标表示?

- 可以表示无穷远点;
- 提供了用矩阵运算把二维、三维甚至高维空间中的一个点集从一个坐标系变换到另一个坐标系的有效方法。

三维空间变换

➤ 空间几何变换:

• 射影变换

- n维射影空间的射影变换 (projective transformation)

$$\rho \mathbf{y} = \mathbf{T}_P \mathbf{x}$$

$$\mathbf{x} = (x_1, x_2, \dots, x_{n+1})^T, \mathbf{y} = (y_1, y_2, \dots, y_{n+1})^T$$

$\mathbf{T}_P$ 为满秩的 $(n+1) \times (n+1)$ 矩阵

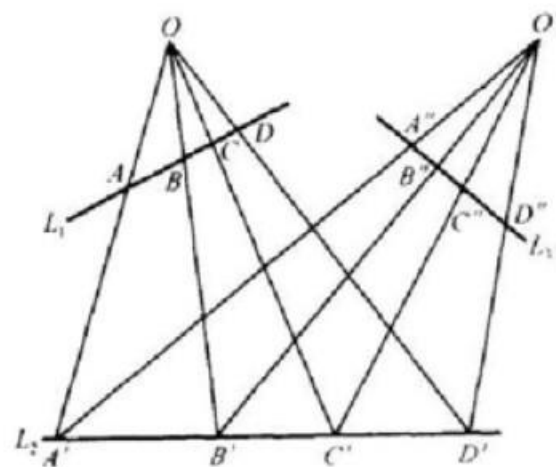
射影变换由 $\mathbf{T}_P$ 矩阵决定, 有 $(n+1)^2$ 个参数,

独立参数 $(n+1)^2 - 1$ 个, 因为 $\mathbf{T}_P$ 与 $k\mathbf{T}_P$ 表示同一变换!

- 1维射影变换:
$$\rho \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\bar{y} = y_1/y_2, \bar{x} = x_1/x_2$$

$$\bar{y} = \frac{m_{11}\bar{x} + m_{12}}{m_{21}\bar{x} + m_{22}}$$



三维空间变换

➤ 空间几何变换:

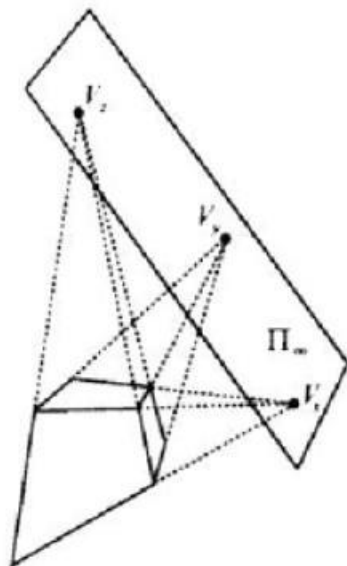
• 射影变换

• n维射影空间的射影变换

• 3维射影变换:

$$\mathbf{T}_P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix}$$

• 16个参数, 15个自由度



三维空间变换

➤ 空间几何变换:

- 仿射变换

- 仿射变换 (affine transformation)

是射影变换的特例，是一类重要的线性几何变换，
当射影中心平面变为**无穷远处**，射影变换变成了仿射变换

- 1维仿射变换:

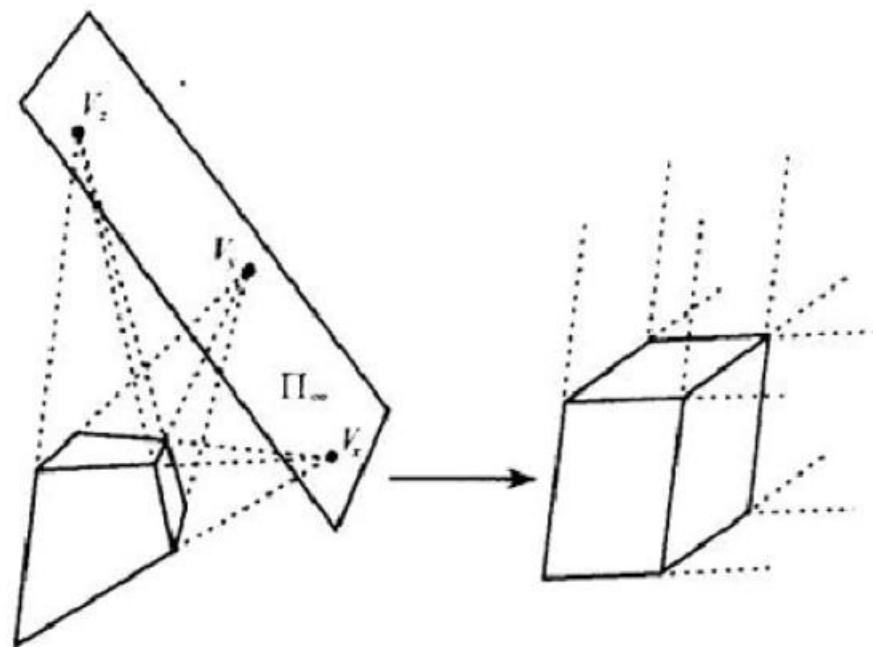
$$\rho \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ 0 & m_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\bar{y} = \frac{m_{11}\bar{x} + m_{12}}{m_{2a}}$$

- 3维仿射变换:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} a_{14} \\ a_{24} \\ a_{34} \end{bmatrix} \quad T_A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 12个自由度



三维空间变换

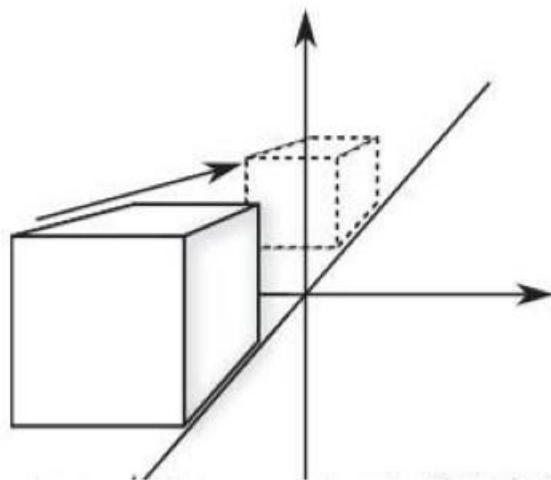
➤ 空间几何变换:

- 比例变换 (相似变换)
- **比例变换 (metric/similarity transformation)**
是带有一比例因子的欧式变换。
比例变换不改变空间物体的形状, 只是改变大小,
因此, 也称为相似变换。
- 3维比例变换:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \rho \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} t_{11} \\ t_{21} \\ t_{31} \end{bmatrix}$$

$$\mathbf{T}_M = \begin{bmatrix} \delta r_{11} & \delta r_{12} & \delta r_{13} & t_{11} \\ \delta r_{21} & \delta r_{22} & \delta r_{23} & t_{21} \\ \delta r_{31} & \delta r_{32} & \delta r_{33} & t_{31} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **7个自由度**
3个旋转
3个平移
1个比例



三维空间变换

➤ 空间几何变换:

- 欧式变换

- 欧式变换 (euclidian transformation)

欧式变换不改变空间物体的形状和大小，只改变位姿。

- 3维欧式变换:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} t_{11} \\ t_{21} \\ t_{31} \end{bmatrix}$$

$$\mathbf{T}_E = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_{11} \\ r_{21} & r_{22} & r_{23} & t_{21} \\ r_{31} & r_{32} & r_{33} & t_{31} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 6个自由度
3个旋转
3个平移

三维空间变换

➤ 空间几何变换:

• 变换关系

$$\mathbf{T}_P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix} \quad \mathbf{T}_A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$\mathbf{T}_M = \begin{bmatrix} \delta r_{11} & \delta r_{12} & \delta r_{13} & t_{11} \\ \delta r_{21} & \delta r_{22} & \delta r_{23} & t_{21} \\ \delta r_{31} & \delta r_{32} & \delta r_{33} & t_{31} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{T}_E = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_{11} \\ r_{21} & r_{22} & r_{23} & t_{21} \\ r_{31} & r_{32} & r_{33} & t_{31} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 仿射变换是射影变换的特例
- 比例变换是仿射变换的特例
- 欧式变换是比例变换的特例

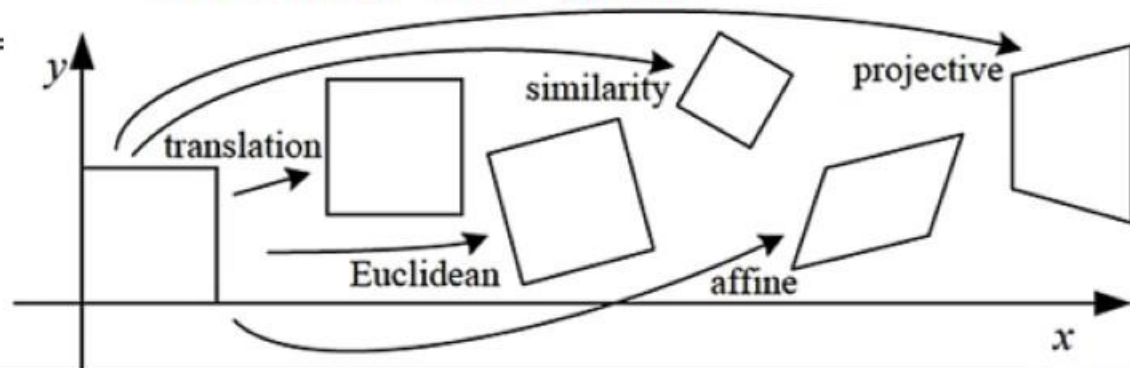
三维空间变换

➤ 空间几何变换:

• 变换关系

Group	Matrix	Distortion	Invariant properties
Projective 8 dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, order of contact : intersection (1 pt contact); tangency (2 pt contact); inflections (3 pt contact with line); tangent discontinuities and cusps. cross ratio (ratio of ratio of lengths).
Affine 6 dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallelism, ratio of areas, ratio of lengths on collinear or parallel lines (e.g. midpoints), linear combinations of vectors (e.g. centroids). The line at infinity, l_∞ .
Similarity 4 dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratio of lengths, angle. The circular points, I, J (see section 2.7.3).
Euclidean 3 dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Length, area

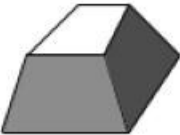
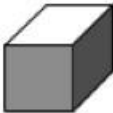
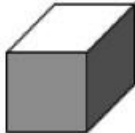
- 接触平面的相交与相切
点线面接触顺序
交比不变
- 平行性, 体积 (面积) 比,
向量的线性组合
- 线和角度的比例, 体积 (面积) 比
- 长度, 夹角, 体积 (面积)



三维空间变换

➤ 空间几何变换:

• 三维变换

Group	Matrix	Distortion	Invariant properties
Projective 15 dof	$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix}$		Intersection and tangency of surfaces in contact. Sign of Gaussian curvature.
Affine 12 dof	$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$		Parallelism of planes, volume ratios, centroids. The plane at infinity, π_∞ , (see section 3.5).
Similarity 7 dof	$\begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$		The absolute conic, Ω_∞ , (see section 3.6).
Euclidean 6 dof	$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$		Volume.

• 旋转、比例、错切、对称

• 平移

$$\mathbf{T}_P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix}$$

• 透视投影

• 总体比例